

Deep Generative Surrogates for Agent-Based Modelling

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Abstract

Agent-Based Models (ABMs) are powerful tools for simulating complex phenomena, but their high computational cost and lack of a tractable likelihood function make parameter inference challenging. This work investigates deep generative surrogates to accelerate Bayesian inference for spatial epidemiological ABMs. We evaluate low-fidelity (MC-Dropout MLP, PriorCVAE) and high-fidelity (diffusion) surrogates on highly optimised simple and complex ABMs. For the simple ABM, a lightweight surrogate accelerated inference by over 100x with comparable predictive performance. For the complex ABM, however, low-fidelity surrogates struggled to accurately model its highly varied behaviour, and the high-fidelity surrogate captured this behaviour better but was slower than the optimised ABM. Furthermore, inference with surrogates for the complex ABM proved difficult, as inaccuracies in their estimation of variance-covariance structure caused the synthetic likelihood to fail. Our findings reveal a critical trade-off between surrogate speed and capacity, and highlight the sensitivity of synthetic likelihoods to surrogate model imperfections.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Timothy James Hitge, 12 June 2025

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1. Introduction

Agent-Based Models (ABMs) or Individual-Based Models (IBMs) are a bottom-up approach to simulating complex phenomena wherein *agents* follow a set of rules determining their interactions with each other and their environment (Jørgensen et al., 2022; Pope and Gimblett, 2015). Whilst there is no canonical definition of the agents in ABMs, a broad definition is that they are independent components of the simulation (Macal and North, 2008). ABMs have been applied in the fields of epidemiology (Jørgensen et al., 2022), economics (Dyer et al., 2024), biology (Colon et al., 2015; Zhang and DeAngelis, 2020) and sociology (Sánchez-Marroño et al., 2015) with agents representing entities such as people, cells, animals, plants and organisations.

An ABM simulation depends on parameters which govern the behaviour of the agents. However, since the appropriate settings for these low-level parameters are not known *a priori*, they must be adjusted so that the ABM's output aligns with observed data or known patterns in the real-world system being modelled (Dyer et al., 2024; Edeling et al., 2021). This process of adjusting ABM parameters is known as *model calibration* (McCulloch et al., 2022). A significant difficulty with calibrating ABMs is that they typically lack a tractable likelihood function, preventing the usage of standard statistical inference techniques for model calibration (Dyer et al., 2024; Jørgensen et al., 2022; McCulloch et al., 2022; Carrella, 2021). This challenge has necessitated the exploration of inference techniques that do not require direct access to the true likelihood function such as Rejection Approximate Bayesian Computation (ABC), Neural Posterior Estimation (NPE), Synthetic or Surrogate Likelihoods, and History Matching as well as Gradient-based Calibration (McCulloch et al., 2022; Dyer et al., 2024; Chopra et al., 2023).

Another central challenge with ABMs is their computational expense; sophisticated ABMs may be very expensive to run, in some cases taking several days for a single forward simulation (Chopra et al., 2023). This computational cost, coupled with the repeated sampling typically required by parameter inference techniques (e.g. Markov Chain Monte Carlo) may render accurate calibration prohibitively expensive. Furthermore, when employing a synthetic likelihood, each evaluation of the likelihood function itself requires that several independent simulations of the ABM are performed.

ABMs have found use in epidemiology due to their effectiveness at incorporating demographic features like population density and mobility data into epidemiological models, as well as for their use in tracing epidemics under different interventions (Harding et al., 2020). During the Covid-19 pandemic, sophisticated ABMs like CovidSim (Ferguson et al., 2025) which depend on tens to hundreds of parameters were developed for the purposes of forecasting and comparing the effectiveness of different intervention strategies to inform policymakers (Ferguson et al., 2020). However, these ABMs were not calibrated in a way that allowed for rigorous uncertainty quantification, leaving policymakers with an incomplete understanding of possible future outcomes (Edeling et al., 2021).

Neural network-based *surrogate models* (or emulators) are a popular approach for approximating complex simulators (Semenova et al., 2025) that have successfully been applied to Agent-Based Models (Jørgensen et al., 2022; Angione et al., 2022). Furthermore, Edeling et al. (2021) note that complex ABMs like CovidSim could benefit from surrogate modelling by developing lightweight surrogates that only use a subset of important ABM parameters. These surrogates could then enable the use of Bayesian inference to obtain posterior distributions over important ABM parameters.

In this work, we develop deep generative surrogates for two epidemiological ABMs which simulate the spread of disease over a spatial grid, with a particular focus on accelerating parameter inference. First, we experiment with the use of PriorCVAE (Semenova et al., 2023) and a Monte Carlo Dropout Neural

Network ([Gal and Ghahramani, 2016](#)) as low-fidelity surrogates for a simple Susceptible-Infected (SI) ABM adapted from [Semenova \(2024\)](#). Next, we extend this ABM to incorporate behaviours such as recovery from the disease, waning immunity and agent movement and compare the effectiveness of the aforementioned low fidelity surrogates with a novel high-fidelity diffusion-based surrogate.

2. Background

We first provide a detailed overview of the core architectures and algorithms that were used in this work. Chapter 3 describes the ABMs that were considered and how the techniques outlined here were employed.

2.1 Monte Carlo Dropout

Modelling uncertainty is crucial for the application of deep learning in many domains, however, deep learning approaches often provide only point estimates of parameters and predictions (Gal, 2016). Bayesian deep learning seeks to combine the power of Bayesian probability theory with modern deep learning architectures to obtain strong performance and model uncertainty (Li and Gal, 2017). Bayesian Neural Networks are a popular form of Bayesian deep learning wherein prior probability distributions are placed over the weights of a Neural Network and Bayesian inference is performed to obtain a posterior distribution over weights (Li and Gal, 2017; Jospin et al., 2022). However, in practice, this is computationally infeasible and as a result a great deal of effort has been devoted to approximate inference techniques based on Markov Chain Monte Carlo and Variational Inference (Jospin et al., 2022).

Gal and Ghahramani (2016) demonstrated that neural networks trained with dropout at every layer can be interpreted as variational approximations of a Bayesian Neural Network with the same network structure. In particular, when a Neural Network is trained with a maximum likelihood objective, Bernoulli dropout noise and ℓ^2 weight penalisation, the training objective is equivalent to maximising the Evidence Lower BOund (ELBO) between a variational posterior and a Bayesian Neural Network with a normal prior on the weights (Jospin et al., 2022). The variational distribution for each $m_i \times n_i$ weight matrix $\mathbf{W}_i$ with probability of dropout p_i is defined as

$$\mathbf{M}_i = \mathbf{W}_i \cdot \text{diag}(\mathbf{z}_i) \quad (2.1.1)$$

$$(\mathbf{z}_i)_j \sim \text{Bernoulli}(p_i) \text{ for } j = 1, \dots, n_i \quad (2.1.2)$$

Once such a Neural Network is trained, performing forward passes with dropout enabled is equivalent to sampling from the approximate posterior and then generating a prediction using those parameters. Hence, the predictive mean and predictive uncertainty can be estimated by performing multiple forward passes on the same inputs but with different dropout noise samples (Gal and Ghahramani, 2016). Furthermore, since these different forward passes can be performed concurrently within a single batch, the running time of evaluating a single example is largely unaffected. The fact that the training procedure is very similar to conventional Neural Networks and that uncertainties are cheap to compute, makes Monte Carlo Dropout (MC-Dropout) a practical way to perform Bayesian Deep learning (Jospin et al., 2022). However, it may underestimate model uncertainty since the underlying variational objective encourages mode-seeking behaviour; the approximate distribution is penalised for placing mass where the posterior *does not* have mass but is not penalised for not placing mass where the posterior *does* have mass (Li and Gal, 2017). Additionally, the dropout rate affects the uncertainty estimates, and not all architectures are amenable to dropout (Kirsch, 2024).

2.2 Variational Auto-Encoder and PriorCVAE

Variational Auto-Encoders (VAE) (Kingma and Welling, 2014) are deep generative models consisting of an encoder network $\mathcal{E}_\phi$ and a decoder network $\mathcal{D}_\theta$ that are jointly optimised using an unsupervised

learning objective. The encoder $\mathcal{E}_\phi$ maps data samples $x \sim p(x)$ to the parameters of a distribution over a *latent* variable $q_\phi(z|x)$ and the decoder $\mathcal{D}_\theta$ directly maps *latent* samples $z \sim p(z)$ into samples from a distribution over reconstructed data $x \sim p_\theta(x|z)$.

In generative modelling, we aim to approximate the underlying data distribution $p(x)$ with a parametrised distribution $p_\theta(x)$. The VAE objective is constructed by incorporating the latent variable z and then deriving an optimisation objective from the Evidence Lower BOund (ELBO) on $p_\theta(x)$. First, $p_\theta(x)$ is decomposed into an integral over all possible values of the latent variable z and multiplied by $\frac{q_\phi(z|x)}{q_\phi(z|x)}$. Hence, we can rewrite $p_\theta(x)$ as an expectation over $q_\phi(z|x)$:

$$p_\theta(x) = \int p_\theta(x, z) \frac{q_\phi(z|x)}{q_\phi(z|x)} dz = \mathbb{E}_{z \sim q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \quad (2.2.1)$$

By Jensen's Inequality we obtain the ELBO:

$$\log p_\theta(x) = \log \mathbb{E}_{z \sim q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \geq \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \quad (2.2.2)$$

Furthermore, since maximising the ELBO maximises $\log p_\theta(x)$, our loss function is the negative ELBO:

$$\mathcal{L}(\theta, \phi, x) = -\mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \quad (2.2.3)$$

$$= -\mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] - \mathbb{E}_{z \sim q_\phi(z|x)} \left[\log \frac{p(z)}{q_\phi(z|x)} \right] \quad (2.2.4)$$

$$= \text{KL}(q_\phi(z|x) \parallel p(z)) - \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] \quad (2.2.5)$$

In the Variational Auto-Encoder, $q_\phi(z|x)$ takes the form of a diagonal Gaussian distribution with parameters $\mu_\phi(x)$ and $\sigma_\phi^2(x)\mathbf{I}$, and $p(z)$, the prior distribution of the latent variable, is typically set to a standard Gaussian distribution. Under this choice of $p(z)$, a well-known analytic solution for the KL-divergence between diagonal Gaussians can be substituted in

$$\mathcal{L}(\theta, \phi, x) = -\frac{1}{2} \sum_{k=1}^K (1 + \log(\sigma_k^2) - \sigma_k^2 - \mu_k^2) - \mathbb{E}_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] \quad (2.2.6)$$

The gradient $\nabla_\theta \mathcal{L}(\theta, \phi, x)$ can easily be computed as $-\mathbb{E}_{z \sim q_\phi(z|x)} [\nabla_\theta \log p_\theta(x, z)]$, however $\nabla_\phi \mathcal{L}(\theta, \phi, x)$ is problematic since the expectation itself is dependent on ϕ . The REINFORCE or score function estimator (Williams, 1992; Murphy, 2023) provides a generic approach to approximate this gradient, however its estimate has very high variance (Kingma and Welling, 2014). Instead, since $q_\phi(z|x)$ is Gaussian, the reparametrisation trick (Kingma and Welling, 2014) can be employed. The reparametrisation trick expresses $z \sim q_\phi(z|x)$ as a deterministic transformation of standard Gaussian noise:

$$z_\phi(x, \epsilon) := \mu_\phi(x) + \sigma_\phi(x) \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (2.2.7)$$

Hence, our loss function becomes:

$$\mathcal{L}(\theta, \phi, x) = -\frac{1}{2} \sum_{k=1}^K (1 + \log(\sigma_k^2) - \sigma_k^2 - \mu_k^2) - \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} [\log p_\theta(x|z_\phi(x, \epsilon))] \quad (2.2.8)$$

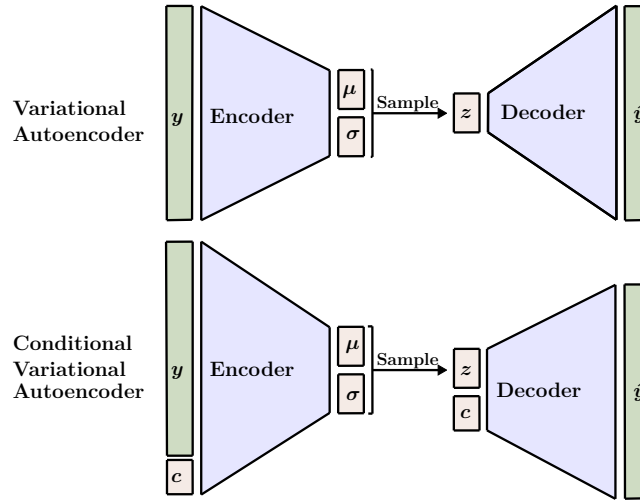


Figure 2.1: The VAE and CVAE architectures from Figure 2 of [Semenova et al. \(2023\)](#).

This reformulation separates the source of randomness from the parameters ϕ , allowing us to bring the gradient operator inside the expectation. Also, note that in practice, the expectation over ϵ is approximated with a single sample. Finally, when the data is continuous, $p_\theta(x|z)$ is assumed to be Gaussian, yielding an MSE term:

$$\mathcal{L}(\theta, \phi, x) = -\frac{1}{2} \sum_{k=1}^K (1 + \log(\sigma_k^2) - \sigma_k^2 - \mu_k^2) + \|x - \mathcal{D}_\theta(z_\phi(x, \epsilon))\|^2, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (2.2.9)$$

VAEs can be extended to incorporate discrete (e.g. a class label) or continuous (e.g. a parameter) conditional information by concatenating a conditioning vector onto the input to the encoder $\mathcal{E}_\phi$ and decoder $\mathcal{D}_\theta$ networks. This is referred to as a conditional VAE (CVAE) ([Sohn et al., 2015](#)) (see Figure 2.1). In our work, we adapted the CVAE model PriorCVAE ([Semenova et al., 2023](#)) to the task of surrogate modelling. PriorCVAE ([Semenova et al., 2023](#)) was proposed in the context of encoding Gaussian process priors for scalable MCMC inference. The PriorCVAE objective incorporates an additional tuning parameter σ_{vae} that alters the quantity of uncertainty learned by PriorCVAE:

$$\mathcal{L}_{\text{PriorCVAE}}(\theta, \phi, x, c) = \frac{1}{2\sigma_{\text{vae}}^2} \|x - \mathcal{D}_\theta(z_\phi(x, c, \epsilon), c)\|^2 - \frac{1}{2} \sum_{k=1}^K (1 + \log(\sigma_k^2) - \sigma_k^2 - \mu_k^2), \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (2.2.10)$$

2.3 Diffusion Models

Diffusion Models ([Sohl-Dickstein et al., 2015](#); [Ho et al., 2020](#); [Song et al., 2021](#)) are a class of deep generative models which iteratively transform samples from a simple distribution into samples from a complex data distribution. The two original diffusion model formulations are the discrete-time Markov chain view of [Ho et al. \(2020\)](#) and [Sohl-Dickstein et al. \(2015\)](#), and the score-based stochastic differential equation (SDE) view of [Song et al. \(2021\)](#). Both perspectives define corruption processes and train

neural networks to learn to reverse these processes. The first perspective defines a corruption procedure wherein Gaussian noise is iteratively added to data until it resembles samples from a standard multivariate Gaussian distribution. This process is referred to as the *forward* or *diffusion* process, and defined by a discrete-time Markov chain with variances $0 < \beta_t < 1$:

$$q(x_{1:T}|x_0) := \prod_{t=1}^T q(x_t|x_{t-1}) \quad (2.3.1)$$

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t \mathbf{I}) \quad (2.3.2)$$

The aim is to learn the transitions of a corresponding *reverse* process which iteratively refines pure Gaussian noise $x_t \sim \mathcal{N}(x_T; 0, \mathbf{I})$ into a sample from the data distribution:

$$p_\theta(x_{0:T}) := p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t) \quad (2.3.3)$$

$$p_\theta(x_{t-1}|x_t) := \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (2.3.4)$$

Typically Σ_θ is set to a time dependent constant, rather than being learnt (Ho et al., 2020; Nichol and Dhariwal, 2021). From the Evidence Lower BOund (ELBO) on the negative log likelihood, an efficient, MSE objective between the means of $q(x_{t-1}|x_t, x_0)$ and $p_\theta(x_{t-1}|x_t)$ can be derived. However, it is commonplace to apply the reparametrisation trick (Kingma and Welling, 2014)

$$x_t(x_0, \epsilon) = \sqrt{\tilde{\alpha}_t}x_0 + \sqrt{1 - \tilde{\alpha}_t}\epsilon, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}) \text{ and } \tilde{\alpha}_t = \prod_{s=1}^t (1 - \beta_s) \quad (2.3.5)$$

and adopt the equivalent ϵ -prediction objective as it often leads to higher quality samples (Ho et al., 2020):

$$\operatorname{argmin}_{\theta} \left\| \epsilon - \epsilon_\theta \left(\sqrt{\tilde{\alpha}_t}x_0 + \sqrt{1 - \tilde{\alpha}_t}\epsilon, t \right) \right\|^2 \quad (2.3.6)$$

The score-based perspective defines a forward *SDE* that similarly corrupts data into Gaussian noise:

$$dx = f(x, t)dt + g(t)d\mathbf{w} \quad (2.3.7)$$

where $\mathbf{w}$ is the standard Wiener process, $f(\cdot, t)$ is the drift coefficient of $x(t)$ and $g(\cdot)$ is the diffusion coefficient of $x(t)$. The connection between the discrete and continuous-time perspectives can be seen by taking the limit $T \rightarrow \infty$ of the discrete time forward process, as this yields the SDE with drift coefficient $-\frac{1}{2}\beta(t)x$ and diffusion coefficient $\sqrt{\beta(t)}$:

$$dx = -\frac{1}{2}\beta(t)xdt + \sqrt{\beta(t)}d\mathbf{w} \quad (2.3.8)$$

Anderson (1982) demonstrated that the reverse of a diffusion process is itself a diffusion process that runs backwards in time and is defined by a reverse-time SDE. The reverse-time SDE for Equation (2.3.7) is

$$dx = \left[f(x, t) - g(t)^2 \nabla_x \log p_t(x) \right] dt + g(t)d\bar{\mathbf{w}} \quad (2.3.9)$$

where $\bar{\mathbf{w}}$ is a standard Wiener process when time flows backwards from T to 0, and dt is an infinitesimal negative timestep. Hence, given the score of the marginal distributions $\nabla_x \log p_t(x)$ for all t , it is possible to simulate the reverse diffusion process to sample from the data distribution $p_0(x)$. This score

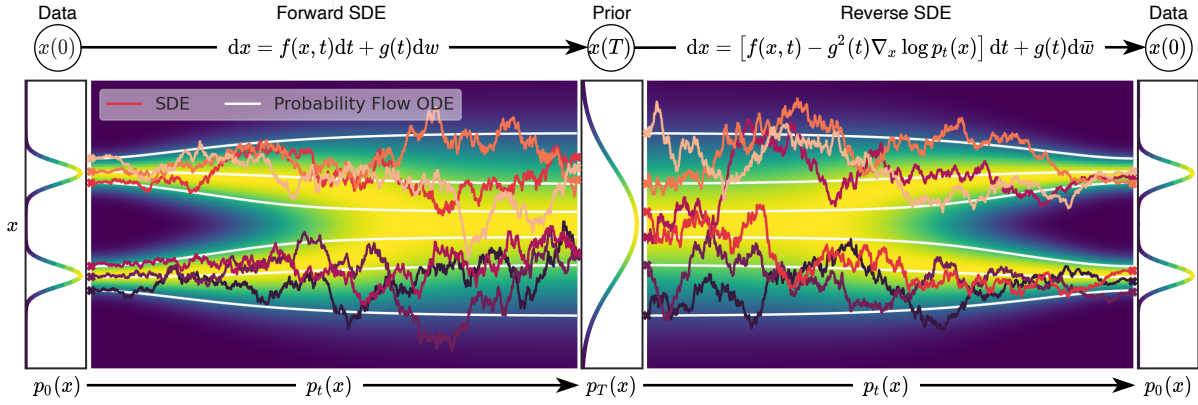


Figure 2.2: Overview of score-based diffusion models taken from Figure 2 of Song et al. (2021). The forward SDE maps data to a noise or prior distribution. Reversing this SDE enables the transformation of samples from the prior into samples from the data distribution. Associated with each SDE is a probability flow Ordinary Differential Equation (ODE) which shares the same marginal probability densities as the SDE. The reverse-time SDE and probability flow ODE can both be obtained by estimating the score $\nabla_x \log p_t(x)$ (Song et al., 2021).

function is estimated at different t through score matching, which similarly reduces to a weighted MSE objective between the neural network's output and the noise ϵ . Furthermore, for each diffusion process, there exists a corresponding deterministic process whose trajectories share the same marginal probability densities as the SDE, and this deterministic process takes the form of an ODE (Song et al., 2021). Therefore, our reverse-time SDE has a corresponding reverse-time *probability flow* ODE:

$$dx = \left[f(x, t) - \frac{1}{2}g(t)^2 \nabla_x \log p_t(x) \right] dt \quad (2.3.10)$$

Hence, instead of solving the reverse-time SDE, we can employ standard ODE solvers when sampling from the diffusion model. Which, in practice, can achieve comparable quality to SDE-based sampling with far fewer solver steps (Karras et al., 2022).

In our experiments with diffusion models, we use a score-based diffusion model based on the so-called *EDM* framework from the seminal work of Karras et al. (2022). They advocate and empirically motivate for a choice of coefficients that yields a simplified reverse-time probability flow

$$dx = -t \nabla_x \log p_t(x) dt \quad (2.3.11)$$

Instead of the neural network directly parametrising a denoiser, the denoised output is given by

$$D_\theta(x; \sigma) = c_{\text{skip}}(\sigma)x + c_{\text{out}}F_\theta(c_{\text{in}}(\sigma)x; c_{\text{noise}}(\sigma)) \quad (2.3.12)$$

F_θ is the raw output of the neural network, c_{in} and c_{out} scale the input and output so that a fixed variance is maintained across different noise levels and c_{noise} is an empirical transformation of the current noise-level. Most importantly, c_{skip} is used to adaptively change the training target between ϵ and x_0 prediction as the noise level varies. When the noise is dominant, c_{skip} is low, and when the clean image is dominant, c_{skip} is high. Therefore, F_θ is forced to mainly predict the denoised image when noise is high and when noise is low it is forced to mainly predict the remaining noise. This is in contrast to the typical x_0 and ϵ prediction objectives which become trivial at low and high noise levels

respectively. The adaptive training objective yields better score function estimates at the beginning of the sampling procedure, enabling the model to produce higher quality samples with fewer denoising steps (Alonso et al., 2024).

2.4 Markov Chain Monte Carlo and Affine Invariant Ensemble Sampler

Bayesian inference is the computation of a posterior distribution $p(\theta|\mathcal{D})$ over variables of interest θ , given data $\mathcal{D}$ and prior beliefs about those variables $p(\theta)$ using Bayes' rule (Murphy, 2023):

$$p(\theta|\mathcal{D}) = \frac{p(\theta)p(\mathcal{D}|\theta)}{p(\mathcal{D})} \quad (2.4.1)$$

However, the computation of the normalisation constant $p(\mathcal{D})$, is often intractable for real-world problems since it entails solving a high dimensional integral:

$$p(\mathcal{D}) = \int p(\mathcal{D}|\theta)p(\theta)d\theta \quad (2.4.2)$$

Monte Carlo approximations of the posterior are unbiased and non-parametric. They are defined in terms of a set of samples $q(\theta) \approx \frac{1}{N} \sum_{i=1}^N \delta(\theta - \theta_i)$ (Murphy, 2023). However, it is necessary to obtain samples from the posterior in order to make this approximation. Markov Chain Monte Carlo (MCMC) is a popular approach to draw samples from a posterior without evaluating the normalisation constant $p(\mathcal{D})$ (Jones and Qin, 2022). MCMC algorithms construct a Markov chain on the parameter space whose stationary distribution is the posterior distribution by performing a random walk over the parameter space so that the number of samples drawn at each point θ_i in the parameter space is proportional to $p(\theta_i|\mathcal{D})$ (Murphy, 2023). The classical MCMC algorithm is the Metropolis-Hastings algorithm (Metropolis et al., 1953), which draws potential new points θ' from a proposal distribution $q(\theta'|\theta)$ and then computes an acceptance probability A which determines the probability of transitioning to that point in parameter space.

$$A = \min(1, \alpha) \quad (2.4.3)$$

$$\alpha = \frac{p(\theta'|\mathcal{D})q(\theta|\theta')}{p(\theta|\mathcal{D})q(\theta'|\theta)} \quad (2.4.4)$$

Importantly, the normalisation constants in α cancel out, meaning that we only need to be able to evaluate the posterior density up to a normalisation constant:

$$\alpha = \frac{p(\theta'|\mathcal{D})q(\theta|\theta')}{p(\theta|\mathcal{D})q(\theta'|\theta)} = \frac{p(\mathcal{D})}{p(\mathcal{D})} \times \frac{p(\theta', \mathcal{D})q(\theta|\theta')}{p(\theta, \mathcal{D})q(\theta'|\theta)} \quad (2.4.5)$$

This procedure defines a Markov chain with a transition matrix:

$$p(\theta'|\theta) = \begin{cases} q(\theta'|\theta)A(\theta'|\theta) & \text{if } \theta' \neq \theta \\ q(\theta|\theta) + \sum_{\theta' \neq \theta} q(\theta'|\theta)(1 - A(\theta'|\theta)) & \text{otherwise} \end{cases} \quad (2.4.6)$$

Therefore, if we choose $q(\theta'|\theta)$ so that the Metropolis-Hastings Markov Chain is *ergodic* and *irreducible* and we use the acceptance ratio to accept or reject proposed moves then the posterior $p(\theta|\mathcal{D})$ is its unique limiting distribution (Murphy, 2023).

¹The summation becomes an integral if the parameter space is continuous.

In our parameter inference experiments, we employed the Affine Invariant Ensemble Sampler (AIES) (Goodman and Weare, 2010; Foreman-Mackey et al., 2013) since it was significantly faster than gradient-based samplers like the No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014). AIES is an MCMC algorithm whose performance is invariant under affine transformations of the parameter space. This makes it well suited to inference problems where parameters are highly correlated. Furthermore, it is easy to parallelise and has few tunable parameters (Foreman-Mackey et al., 2013). AIES utilises an ensemble of K walkers $\{W_i\}_{i=1}^K$ in the parameter space, where the proposal distribution for each walker is based on the current positions of the other walkers. To update the position of walker W_i , a walker W_j is randomly selected from the other walkers and a new position is proposed via an affine transformation

$$W'_i = W_j + Z(W_i - W_j) \quad (2.4.7)$$

$$z \sim g(z), \quad g(z) \propto \begin{cases} \frac{1}{\sqrt{z}} & \text{if } z \in \left[\frac{1}{a}, a\right] \\ 0 & \text{otherwise} \end{cases} \quad (2.4.8)$$

In Equation (2.4.8), a is an adjustable scale parameter typically set to 2. AIES has found widespread adoption in the astrophysics community (Collaboration, 2022) and has also been applied in fields such as geophysics (Issan et al., 2023) and biology (Linden et al., 2022).

3. Methods

Although numerous ABMs have been created, finding suitable ABMs for our purposes proved difficult. Firstly, many ABMs are written in the domain specific language NetLogo (Wilensky, 1999), making it cumbersome to integrate them into machine learning pipelines. Furthermore, we made use of Numpyro (Phan et al., 2019; Bingham et al., 2019) for accelerated parameter inference, restricting us to ABMs that can be traced and *Just-In-Time* (JIT) compiled by the accelerated computing framework JAX (Bradbury et al., 2018). We therefore improved one existing JAX implementation of a spatial Susceptible-Infected ABM (Semenova, 2024) and developed a new spatial Susceptible-Infected-Recovered ABM which is significantly more complex. Implementing these ABMs in JAX allows them to be run on GPUs and benefit from JAX’s powerful composable transformations; compilation via `jax.jit`, gradient computation via `jax.grad` and automatic vectorisation and parallelisation with `jax.vmap` and `jax.pmap` respectively. Furthermore, improving the performance of an ABM may entirely eliminate the need for a surrogate model. Therefore, as a first step ABMs should be made to properly utilise computer hardware (Semenova et al., 2025). Hence, performance comparisons between our surrogates and ABMs are appropriate as they are highly optimised.

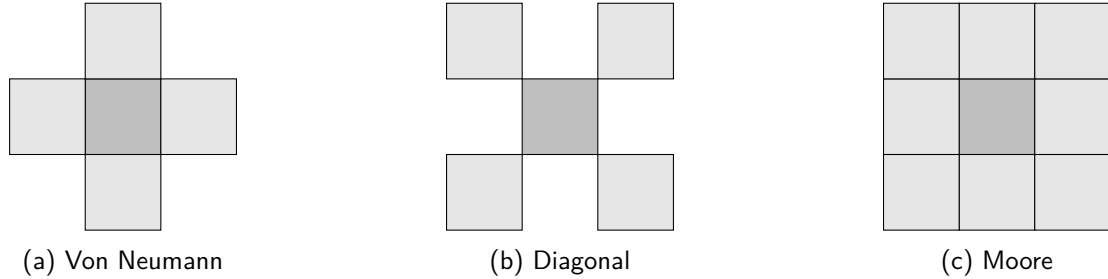


Figure 3.1: Common spatial ABM cell neighbourhoods

Below we elaborate on the two ABMs that we developed, as well as the surrogates we trained for those ABMs and our approach to parameter inference.

3.1 Susceptible-Infected (SI) ABM

3.1.1 SI ABM Implementation

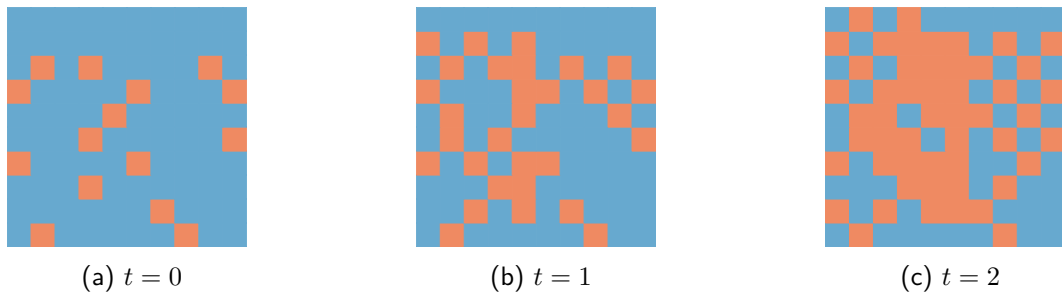


Figure 3.2: A simulation of the SI ABM with a diagonal neighbourhood.

As a first step toward developing efficient surrogates for ABMs, we experiment with a simplistic grid-based model of disease spread. The Susceptible-Infected (SI) ABM, extends the basic SI compartmental

model to a two-dimensional grid. Each cell (agent) in the grid is susceptible or infected, and infection spreads from an infected cell to a susceptible neighbour with probability p_{infect} . A cell's neighbours are defined by a neighbourhood matrix (see Figure 3.1) which defines the spatial interaction pattern for the grid. We adapt an implementation of this ABM which uses a diagonal neighbourhood from [Semenova \(2024\)](#), where the state of the ABM is represented with a binary grid reserving 1 for infected cells and 0 for susceptible cells. Although this original version was implemented in JAX ([Bradbury et al., 2018](#)), we were able to substantially improve its performance by replacing its complex grid update logic with two-dimensional convolutions and elementwise comparisons between matrices. Rather than looping over each susceptible agent's infected neighbours and conducting independent Bernoulli trials to determine if the agent becomes infected, we instead perform a 2D convolution between the binary grid and a binary 3x3 matrix representing the diagonal neighbourhood. This operation counts the number of infected cells in each cell's neighbourhood. Furthermore, the effective probability of infection for a susceptible cell depends only on its number of infected neighbours n_{adj} :

$$p_{\text{effective}} = 1 - (1 - p_{\text{infect}})^{n_{\text{adj}}} \quad (3.1.1)$$

Hence, we can perform a grid update with a single convolution followed by conducting a single Bernoulli trial at each grid position. These changes led to a speedup of over 620x at the largest grid size we considered (Figure 3.3).

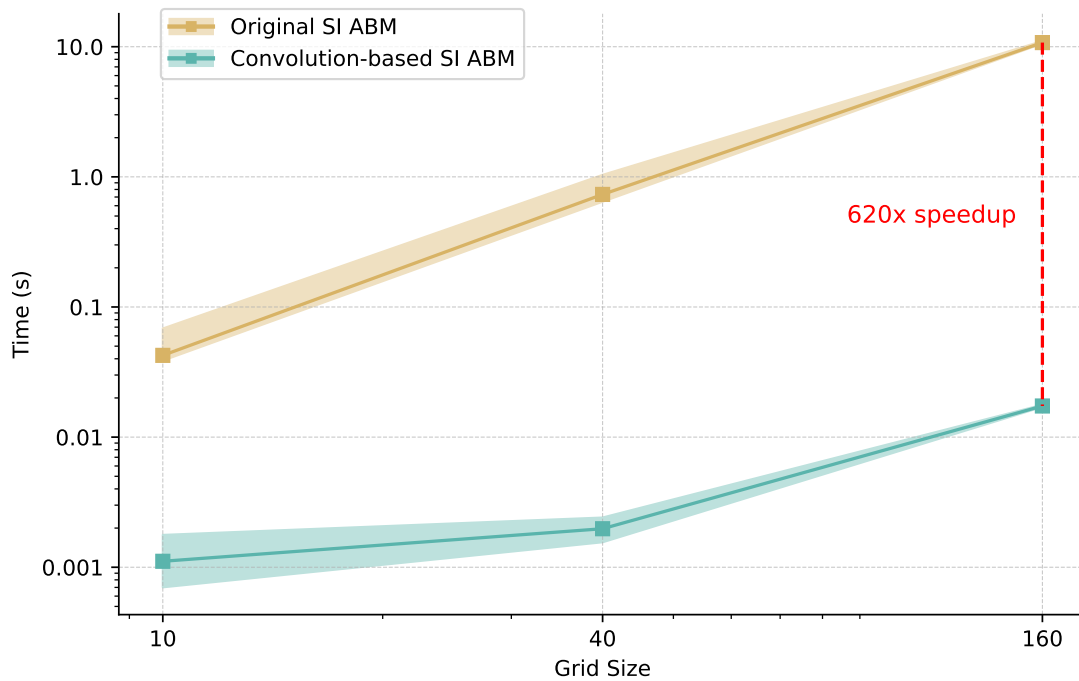


Figure 3.3: Log-log plot comparing the time taken to produce 50 realisations for the convolution-based SI ABM implementation with the original. Solid lines indicate the mean across 50 timings on a NVIDIA® GeForce® MX550 GPU and the shaded regions represent the 95% interpercentile range.

A key difference between this SI ABM and the traditional SI compartmental model is that the ABM is stochastic. Furthermore, its aleatoric uncertainty depends both on the parameters of the ABM and

the size of its grid. Larger grid sizes correspond to larger populations, leading to nearly deterministic behaviour when the grid size grows very large. Figure 3.4 exemplifies this behaviour; we observe a clear trend of larger grid sizes exhibiting less stochasticity across all the parameter settings. Additionally, at smaller grid sizes a larger proportion of the grid cells form its edges. Disease spreads slower to and from these edge cells, since each edge cell necessarily has fewer neighbours. This makes the behaviour of smaller grids more sensitive to the initial spatial distribution of infected agents.

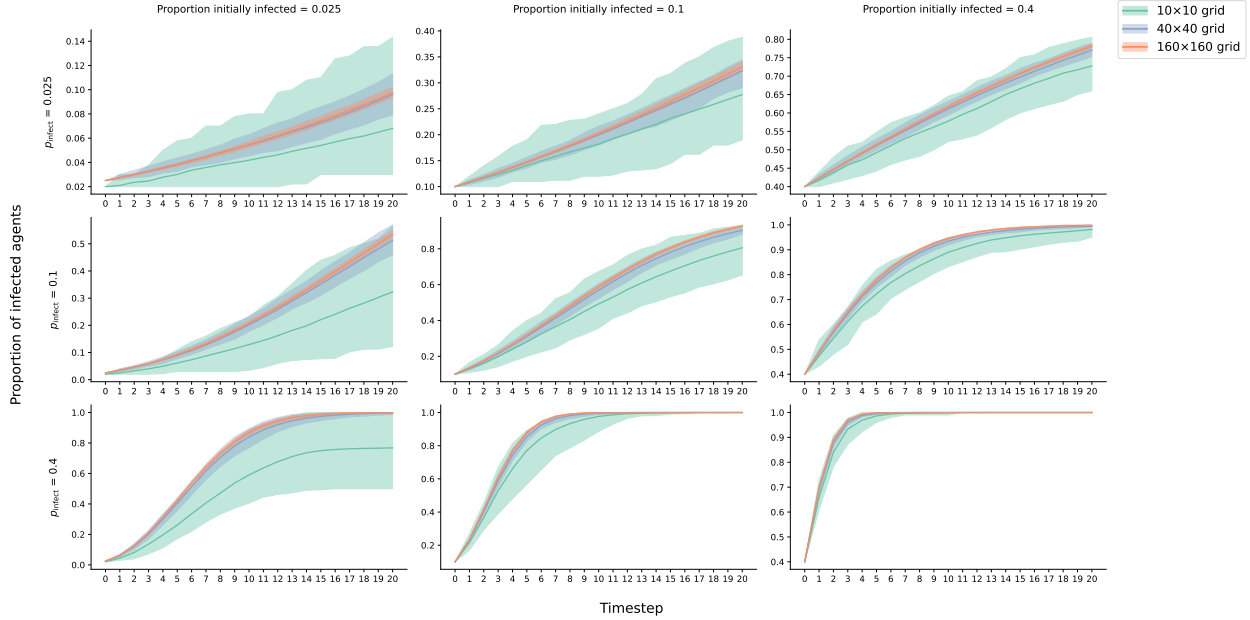


Figure 3.4: The stochasticity of the infection count realisations of the SI model with a diagonal neighbourhood as the probability of infection, size of the initial infected population and grid size vary. The solid lines depict the mean realisation and the shaded regions depict the 95% interpercentile range calculated over 50 independent seeds.

3.1.2 SI Surrogates

We investigated two different classes of surrogate for this ABM; a Monte Carlo Dropout Multi-Layer Perceptron (MCMLP) (Section 2.1) and PriorCVAE (Section 2.2). Rather than directly modelling the spatial evolution of the ABM, these surrogates were trained on the infection-count realisations derived from the evolution of the ABM simulation, significantly lowering the dimensionality of the data. We generated six datasets across three ABM grid sizes; 10×10 , 40×40 and 160×160 , as well as across 10 and 20 ABM simulation steps. The datasets were generated by grid-sampling of p_{infect} and the proportion of initially infected individuals, and sampling 20 independent infection time-series from the ABM per-parameter pair. Hyperparameter sweeps were performed for both surrogates across all six datasets using the Weights and Biases (Biewald, 2020) Bayesian optimisation sweeper with a budget of 100 training runs per surrogate (See Table A.1). Finally, the surrogates were retrained with their best performing configurations on both the tune and train set.

3.1.3 Parameter inference with a surrogate likelihood

A significant difficulty with performing parameter inference for ABMs is that they typically lack a known or tractable likelihood function. As a result, inference must be performed either with likelihood-

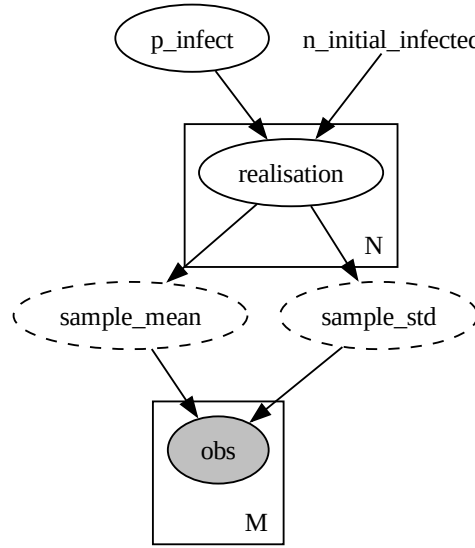


Figure 3.5: A graphical representation of the Bayesian hierarchical model for performing parameter inference on a stochastic SI ABM using NumPyro (Phan et al., 2019; Bingham et al., 2019). p_{infect} is sampled from $\text{Beta}(5, 2)$, $n_{\text{initial_infected}}$ is a fixed variable, N realisations are drawn from the ABM or surrogate, the sample mean and sample standard deviation are calculated for each timestep and the likelihood is given by a diagonal Normal distribution.

free techniques such as the Approximate Bayesian Computation (ABC) rejection algorithm, or with a synthetic (or surrogate) likelihood function (Wood, 2010). A related challenge is that the form of the real-world data used for parameter inference may substantially differ from the form of observations produced by the ABM. In our case, whilst the ABM's simulation proceeds on a grid, our *mock* real-world data will take the form of timeseries of state counts (e.g. infected) over time, since this data is more likely to be readily available and it simplifies our inference problem. We adopt a synthetic likelihood function based on Wood (2010), which approximates the likelihood of summary statistics using a Gaussian distribution whose parameters are estimated by running simulations from the computational model (in this case an ABM) at a given parameter level θ . Concretely, given a vector of observations $\mathbf{x}$ and a parameter sample θ , we perform N independent simulations of the ABM with θ , compute timeseries of state counts $\{y_i\}_{i=1}^T$ and calculate the mean μ_{θ_i} and variance σ_{θ_i} of each y_i across the simulations. Our likelihood function θ is then given by $\mathcal{N}(\mu_{\theta}, \sigma_{\theta} \mathbf{I}) = \prod_{i=1}^T \mathcal{N}(\mu_{\theta_i}, \sigma_{\theta_i})$. Whilst Wood (2010) estimated a full covariance matrix for the likelihood, we decided to make an independence assumption to reduce the computational overhead of inference since we conducted 1800 inference experiments. In practice, we found that even with the independence assumption the ABM was generally able to recover the underlying parameters (Appendix A.2.1). We used the Beta prior from Semenova (2024) for $p(p_{\text{infect}})$. Our Bayesian hierarchical model is depicted in Figure 3.5 where `sample_mean` and `sample_std` correspond to μ_{θ} and σ_{θ} .

Whilst the No U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014) is a common choice for MCMC parameter inference with NumPyro, our early experiments showed that it was significantly slower than

AIES (Section 2.4). This is likely due to NUTS's reliance on computing $\nabla_{\theta} \log p(\theta, D)$ which requires differentiating through the ABM simulation, whilst AIES is gradient-free.

3.2 Susceptible-Infected-Recovered (SIR) ABM

3.2.1 SIR ABM Implementation

In order to better understand the challenges associated with complex real-world ABMs, we developed a generalised SIR ABM. Similar to the classical SIR ODE model, we have susceptible, infected and recovered agents. Recovered agents cannot be infected, however, we allow for the possibility of a recovered agent's immunity to wane, making them susceptible again. The infected agents recover at each timestep with probability p_{recover} and recovered agents become susceptible again with probability p_{wane} . Furthermore, the ABM may now be partially occupied and the random movement of agents into unoccupied cells can be enabled. First, the update rules are applied for the spread of infection, recovery of already infected agents, and the loss of immunity from recovery; afterward agents will attempt to move into a cell in their neighbourhood with probability p_{move} . A move is valid if the cell is unoccupied and tie-breaks between multiple agents attempting to move into the same cell are handled by randomised prioritisation. Figure 3.6 depicts a simulation of the SIR ABM on a 40×40 grid with a total population of 40.

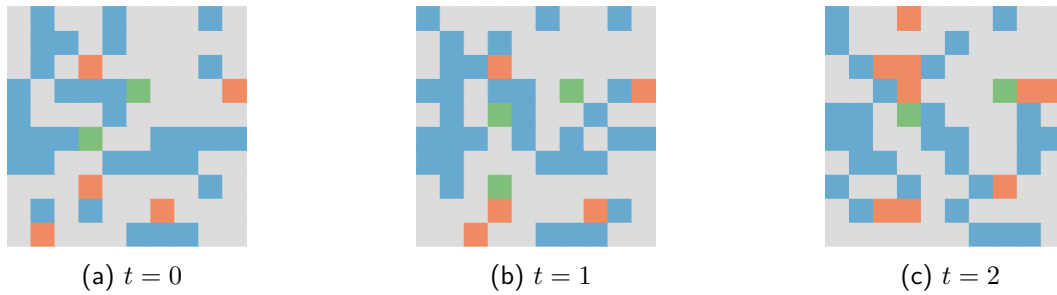


Figure 3.6: A simulation of the SIR ABM with a Von Neumann neighbourhood. Blue cells represent susceptible agents, orange cells represent infected agents, green cells represent recovered agents and gray cells are unoccupied.

The update logic for this ABM is substantially more complicated than the SI ABM largely due to the possibility of agent movement. However, we were able to maintain compatibility with JAX's (Bradbury et al., 2018) JIT-compilation. With fully-static arguments, we are able to sample a batch of 20 realisations from a 40×40 grid run for 50 steps in under 0.2 seconds on a NVIDIA® GeForce® MX550 GPU.

The addition of agent movement and control over agent density significantly augments the sophistication of the ABM. In Figure 3.7, the effect of varying these parameters whilst keeping the others fixed is depicted. We observe that simply changing one of these parameters can lead to an enormous change in the behaviour of the ABM. For example, increasing the probability of agent movement can lead to a much higher peak of infected agents or even lead to an outbreak where there was none. Furthermore, it is possible to run the simulation up to a point with an initial p_{move} and then resume the simulation with a different p_{move} . This potentially allows the SIR ABM to be used to model the impact of social-distancing interventions.

Due to the increased computational demands of performing parameter inference with this ABM and

training high fidelity surrogates, we performed our experiments on an NVIDIA[®] L4 GPU.

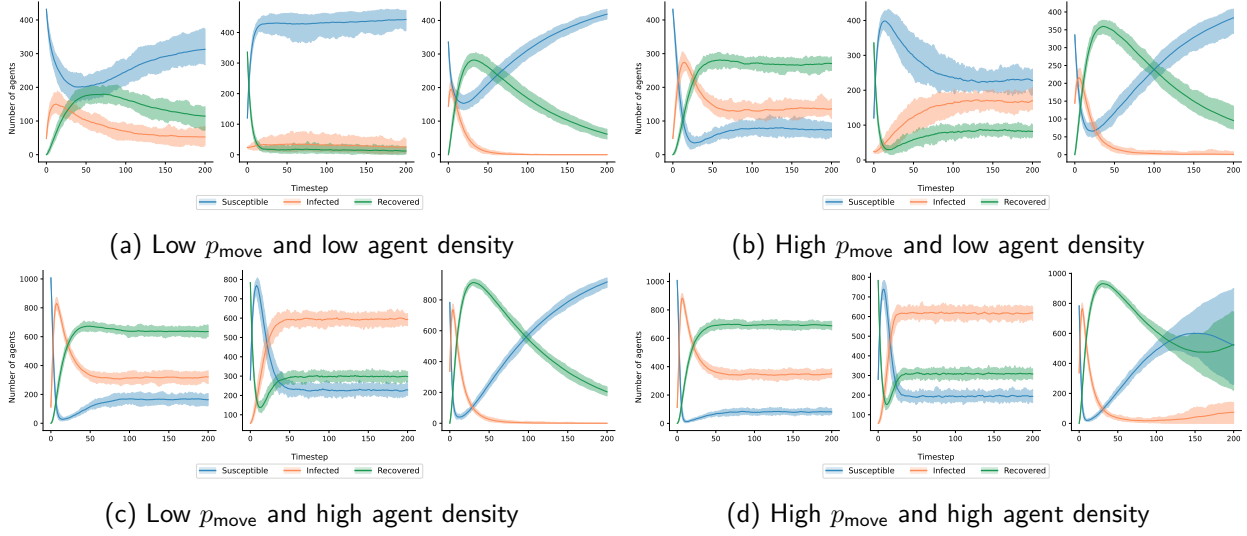


Figure 3.7: The effect of varying p_{move} and the total population of agents in a 40×40 grid¹.

3.2.2 Diffusion-based SIR Surrogates

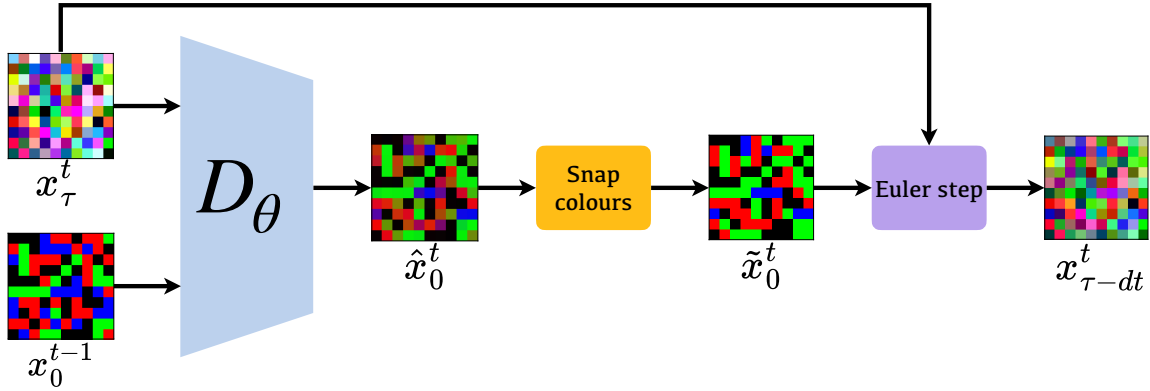


Figure 3.8: A single diffusion step with colour snapping before the Euler update. This denoiser D_θ conditions on a single previous grid x_0^{t-1} ; the conditioning on diffusion time τ and the ABM parameters are not depicted. The previous step in the reverse diffusion process is x_τ^t , $\hat{x}_0^t$ is the denoised prediction, $\tilde{x}_0^t$ is the snapped denoised prediction and $x_{\tau-dt}^t$ is the next step in the reverse diffusion process.

Whilst surrogates that model summary statistics of realisations from the ABM may prove sufficient in simpler cases, these surrogates may require an enormous number of samples to fully capture the complexity present in this ABM as they are not able to directly model agent-level interactions. This motivates the need for surrogates which can fully represent the internal state of the ABM and directly learn its update rules. Drawing inspiration from work on diffusion-based world models (Alonso et al., 2024), we experimented with a diffusion model surrogate for the SIR ABM. Alonso et al. (2024) developed

¹ABM parameters from left-to-right; ($p_{\text{infect}} = 0.6$, $p_{\text{recover}} = 0.05$, $p_{\text{wane}} = 0.025$, $i = 0.1$, $r = 0.0$), ($p_{\text{infect}} = 0.3$, $p_{\text{recover}} = 0.1$, $p_{\text{wane}} = 0.2$, $i = 0.05$, $r = 0.7$) and ($p_{\text{infect}} = 0.4$, $p_{\text{recover}} = 0.1$, $p_{\text{wane}} = 0.01$, $i = 0.3$, $r = 0.0$)

a framework for training reinforcement learning agents inside a diffusion-based world model of Atari video games (Kaiser et al., 2020). They adopted the EDM diffusion framework (Karras et al., 2022) and their denoiser is a time-conditioned U-Net (Ronneberger et al., 2015). The diffusion world model operates autoregressively; it predicts one frame at a time, conditioning on previously generated frames and the agent’s action at the current timestep. Conditioning is achieved via frame-stacking the previous frames (*i.e* concatenating channel-wise) and adding embeddings of the agent’s (discrete) actions to the timestep conditioning. We adapted their architecture to the ABM setting where our analogue of agent actions are ABM parameters. Since our ABM parameters are continuous we obtain features for them via a small dense layer, but otherwise the conditioning mechanism is unchanged. We one-hot encode our ABM grid so that red represents infected, blue represents susceptible and green represents recovered.

Furthermore, we greatly simplified the training procedure, since we have a static dataset and are not training a reinforcement learning agent. Our dataset consists of realisations from the ABM stored as sequences of images. During training, a batch is formed by randomly selecting realisations and then selecting a random segment of each realisation. The segment’s length is determined by the number of previous frames used for conditioning and the number of autoregressive steps performed during training. With no autoregressive steps, the loss is computed by predicting the next frame conditioned on the previous frames sampled from the ABM. When training with autoregressive steps, the loss for the batch is computed by autoregressively predicting a continuation of the segment and summing the intermediate losses. This trains the model on its own outputs which may reduce *exposure bias* (Ranzato et al., 2016) compared to always conditioning on the ground truth frames. Alonso et al. (2024) trained their diffusion models with 1 autoregressive step, however we found it useful to increase this number to 5.

Our reverse diffusion sampler used three denoising steps, the Euler ODE solver and the default diffusion sampling settings used by Alonso et al. (2024). However, instead of conditioning on the previous four frames, we only conditioned on one. The reason is twofold; firstly, conditioning on four frames would require us to sample the first four frame of our simulation from the ABM when evaluating the diffusion model, potentially making the modelling task easier since very large changes often occur within the first few steps, and secondly, because the ABM has the Markov Property, it is only necessary to observe the previous state to be able to accurately model the distribution over the next states.

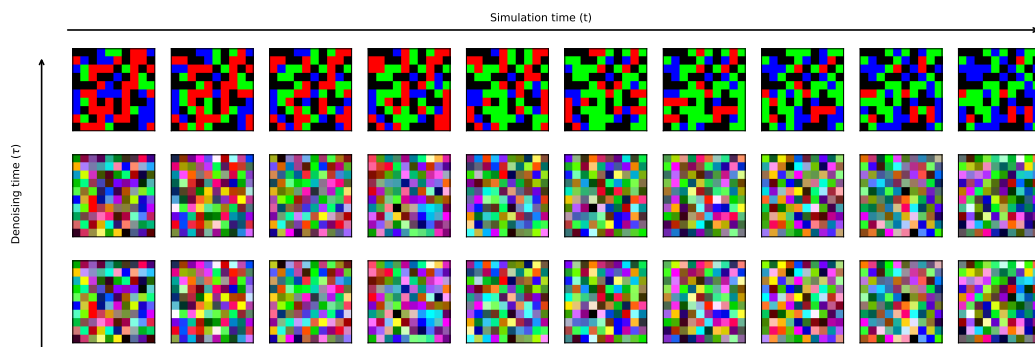


Figure 3.9: The sampling of a realisation from the diffusion surrogate. Producing a single frame requires three diffusion sampling steps conditioned on the previous frame.

Unlike the video game environments the distribution over pixel values in the ABM frames is categorical. This makes it somewhat unsuited to diffusion which is continuous by nature. Furthermore, since the surrogate must condition on its previous outputs, errors compound over time. We developed a mitigation for this problem where the pixel values of the denoised predictions during each diffusion step are first

snapped to their nearest valid colour before the Euler update (Figure 3.8). This has the effect of ensuring that the denoised prediction with which we update our intermediate diffusion state with is always “clean”, in the sense that all of the pixels are meaningful. In Figure 3.9, we depict the generation of a realisation with the diffusion surrogate. A new grid state is produced via diffusion sampling with three denoising steps conditioned on the previous grid state and this process is repeated as we generate the realisation. We use the notation x_τ^t to refer to a grid that is at *simulation* timestep t and *diffusion* time τ . Also note that diffusion time runs backwards from noise to denoised, hence, $\tau = 0$ represents the final prediction of the diffusion model, which corresponds to the top row in Figure 3.9.

3.2.3 Adjustments to the surrogate likelihood

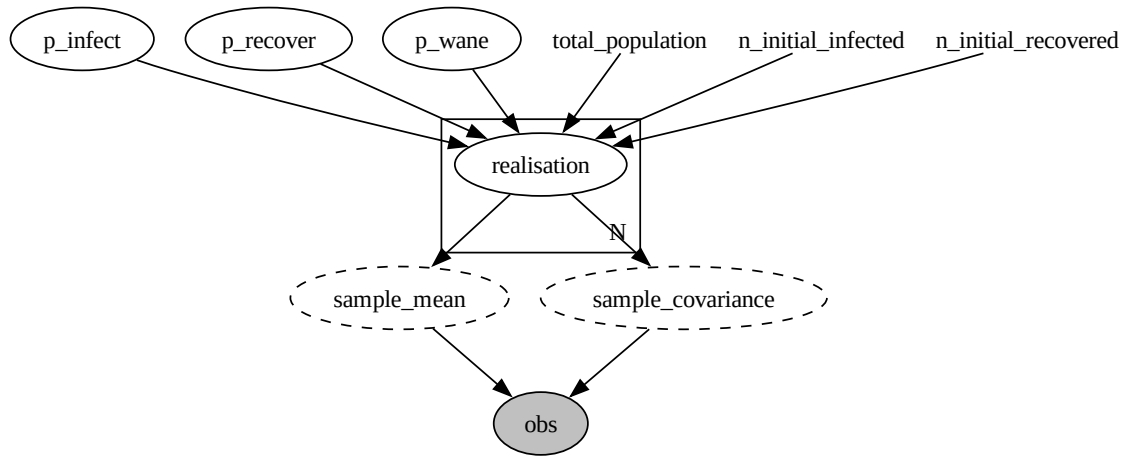


Figure 3.10: A graphical representation of the Bayesian hierarchical model for performing parameter inference on a stochastic SIR ABM using NumPyro. p_{infect} , p_{recover} and p_{wane} are sampled from $\text{Beta}(5, 2)$, and total_population , $n_{\text{initial_infected}}$ and $n_{\text{initial_recovered}}$ are fixed variables. N realisations are drawn from the ABM and the sample mean and sample covariance of a combined infection and recovery timeseries are used as the parameters of the synthetic likelihood.

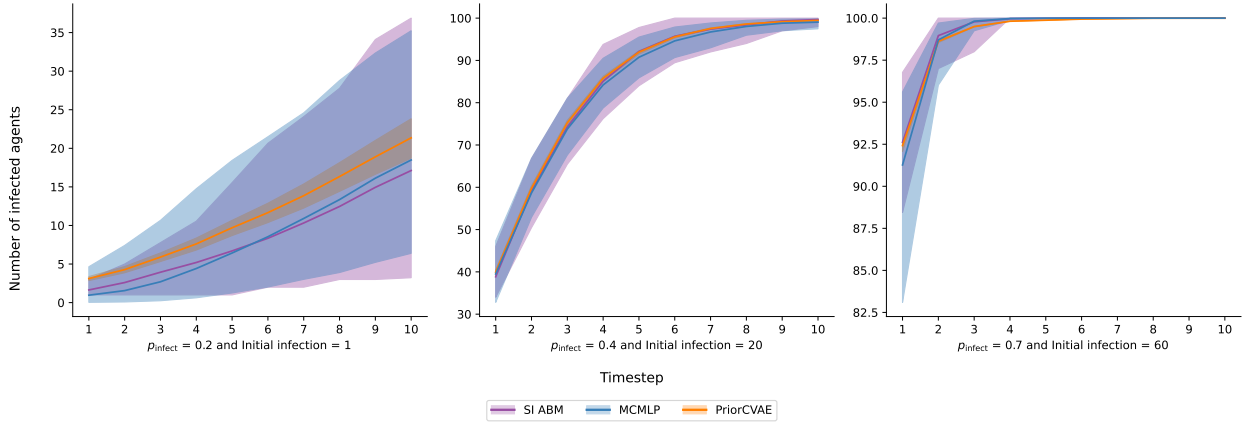
When performing parameter inference for the SIR ABM, we found it necessary to estimate the full covariance of the realisations instead of making the independence assumption. Since the sum of the Susceptible, Infected and Recovered counts at each timestep are equal to the constant total population, estimating the covariance matrix over all three timeseries would result in a singular covariance matrix. We therefore calculate the mean and covariance of a combined infection and recovery timeseries and discard the susceptible timeseries.

In contrast, we found that the MCMLP and PriorCVAE surrogates could not use this full covariance likelihood for parameter inference. Instead, we used a fixed diagonal covariance and estimated the mean of the all timeseries. The reasons for this difference are detailed in Section 4.2.1.

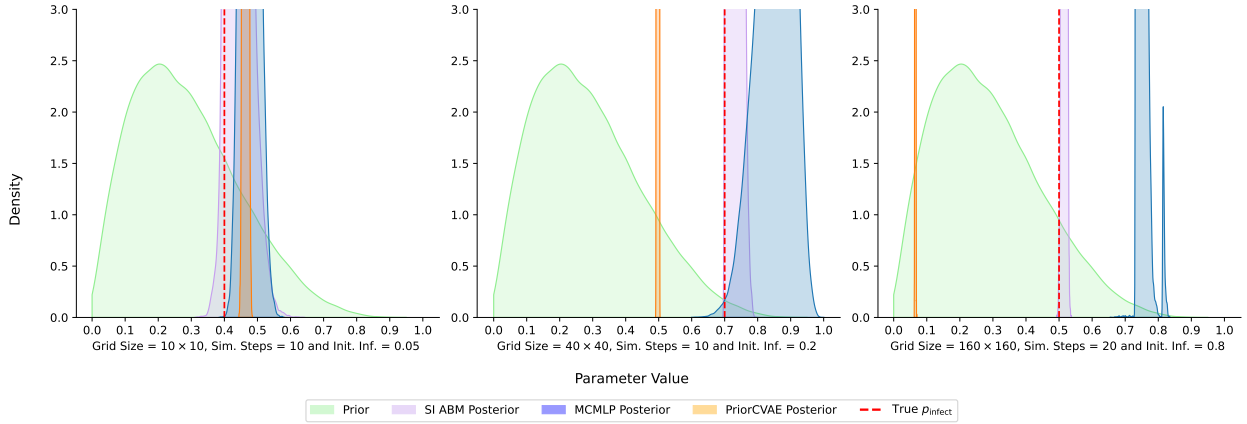
4. Results

4.1 Susceptible Infected Spatial ABM

A visual inspection of the realisations produced by the surrogates revealed that PriorCVAE consistently underestimates the amount of variability at each parameter setting. This deficiency is more apparent at smaller grid sizes where the ABM exhibits greater stochasticity (Figure 4.1a). Additionally, the posterior distributions obtained from parameter inference with PriorCVAE are typically much more concentrated than the ABM posteriors, in many cases collapsing into point masses (Figure 4.1b and Appendix A.2.1).



(a) Variability of surrogate realisations



(b) Surrogate parameter uncertainty

Figure 4.1: (a): Mean and 95% interpercentile ranges of realisations drawn from the ABM and surrogates for a 10×10 grid. (b): Parameter inference examples where PriorCVAE's posterior is far more concentrated than the ABM's posterior.

This is a natural consequence of underestimating the degree to which realisations from a given parameter vary, since the variances of our synthetic likelihood function are calculated from samples drawn from the surrogate. In contrast, the MCMLP surrogates rarely underestimate the variability of realisations or parameter uncertainty, and are more prone to overestimate these quantities. These differences in *parameter* uncertainty also impact the *predictive* uncertainty of the surrogates where we again observe

PriorCVAE’s tendency to underestimate, and MCMLP’s tendency to overestimate (Appendix A.2.1).

In Appendix A.2.1, we present all the inferred posteriors and posterior predictions across different combinations of p_{infect} and proportion of individuals initially infected. These inference experiments revealed the practical difficulties associated with scaling even simple ABMs. The wall-clock time of parameter inference with the ABM at a 160×160 grid size became so long that it was necessary to decrease the number of test cases from 100 to 9. In Figure 4.2, the time taken for parameter inference with the ABM grows to be two-orders of magnitude greater than the surrogates at the largest grid size. These results are bolstered by the fact that training these surrogates takes less than a minute on a NVIDIA® T4 GPU; if a dataset of realisations from the ABM is readily available, it may be orders of magnitude faster to first train a surrogate and then perform parameter inference than to perform inference with the ABM directly. Furthermore, over the course of an iterative Bayesian workflow (Gelman et al., 2020), parameter inference would be performed multiple times, amortising the cost of training the surrogate. An additional advantage offered by surrogates of this form is that they predict entire realisations in a single forward pass, allowing computational cost to stay nearly constant as the sequence length of the projections increase. However, it should be noted that since the next state of the grid (or next infection count) depends only on the last state, the MCMLP and PriorCVAE surrogates can generate longer sequences than they were trained for. This can be achieved by setting the initial infection input to the proportion that were infected in the last timestep of the previous generation.

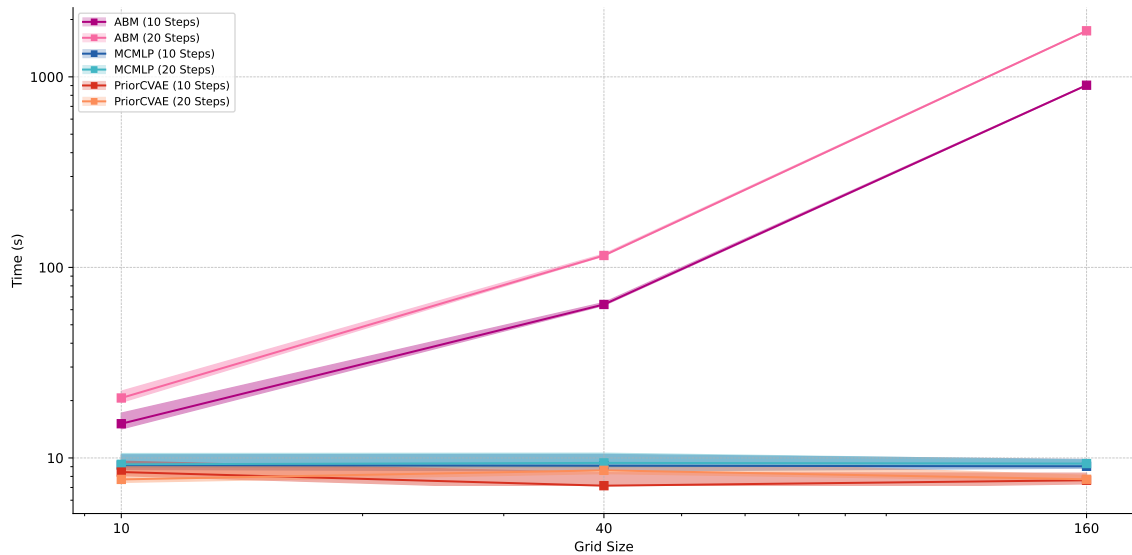


Figure 4.2: Log-log plot of the time taken to perform parameter inference with simulated SI data on a NVIDIA® T4 GPU. The solid lines indicate mean and the shaded regions indicate 95% interpercentile ranges.

The results in Table 4.1 demonstrate that the MCMLP surrogates’ posterior means are on average close to the ABM’s. We also calculated the 1-Wasserstein (W1) or Earth Mover’s Distance between the surrogate and ABM posteriors. The W1 distance, measures the minimum ‘work’ required to transform one distribution into another. In the case of empirical distributions, $\alpha = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}$, if the points are ordered, the W1 distance between α and β reduces to an average over pairwise absolute distances (Peyré and Cuturi, 2020) as seen in Equation (4.1.1).

$$W_1(\alpha, \beta) = \frac{1}{n} \sum_{i=1}^n |x_i - y_i| \quad (4.1.1)$$

The mean 1-Wasserstein distances between the surrogate and ABM posteriors, indicate that not only are the MCMLP surrogates better able to match the posterior mean of the ABM, but that their overall distributions are more similar to the ABMs' than PriorCVAE.

Table 4.1: Mean Squared Error (MSE) and Mean 1-Wasserstein (W_1) between the SI ABM and the Surrogate posteriors. The MSE metric is calculated between the posterior means.

Grid Size	Num. Steps	MSE		Mean W_1	
		MCMLP	PriorCVAE	MCMLP	PriorCVAE
10	10	0.006	0.067	0.050	0.160
	20	0.010	0.136	0.065	0.257
40	10	0.004	0.066	0.042	0.155
	20	0.022	0.098	0.109	0.208
160	10	0.007	0.127	0.067	0.258
	20	0.018	0.162	0.101	0.280

In order to evaluate the predictive performance of the surrogate models we computed the Negative Log Predictive Density (NLPD) for $\{x_i\}_{i=1}^{1000}$ groundtruth realisations generated by the ABM for each of the parameters θ that we performed inference for. We use $p(\cdot|D)$ to denote the posterior predictive probability density function obtained from the parameter inference procedure with data D . In our case this function was estimated from posterior predictive samples using a Gaussian Kernel Density Estimator (Rosenblatt, 1956; Parzen, 1962).

$$\text{NLPD} := -\frac{1}{N} \sum_{i=1}^N \log p(X = x_i|D) \quad (4.1.2)$$

The NLPD is a proper scoring rule that is widely used for evaluating the predictive performance of probabilistic models (McCutcheon, 2019). The NLPD also has an interpretation as an empirical estimate of the cross entropy between the true data generating distribution and the predictive posterior distribution (Oosterheld et al., 2023). If p^* is our true distribution, then Equation (4.1.2) is an estimate of

$$\mathbb{E}_{x \sim p^*(x)} [-\log p(x|D)] = H(p^*) + \text{KL}(p^* || p) \quad (4.1.3)$$

It follows, that when we compare the NLPDs of two distributions with the same true data distribution p^* , the distribution with a lower NLPD, has a lower (estimated) KL-divergence from p^* .

In Table 4.2 we present the number of inference test cases for which each surrogate achieved an NLPD that was *at least* as low as the ABM's. These results indicate that with the exact same MCMC configuration, the MCMLP surrogates are able to obtain an NLPD at least as low as the ABM's in half of the test cases across the different ABM settings. This is promising in light of their superior run time performance (Figure 4.2), since these results may be improved by performing inference with a greater sampling budget and number of chains, a different inference algorithm or prior.

Table 4.2: Count of NLPDs that were at least as low as the ABM's

Grid Size	Num. Steps	MCMLP	PriorCVAE
10	10	46/100	15/100
	20	50/100	31/100
40	10	79/100	8/100
	20	51/100	12/100
160	10	8/9	6/9
	20	7/9	7/9

4.2 Susceptible Infected Recovered Spatial ABM

4.2.1 Diffusion and MCMLP

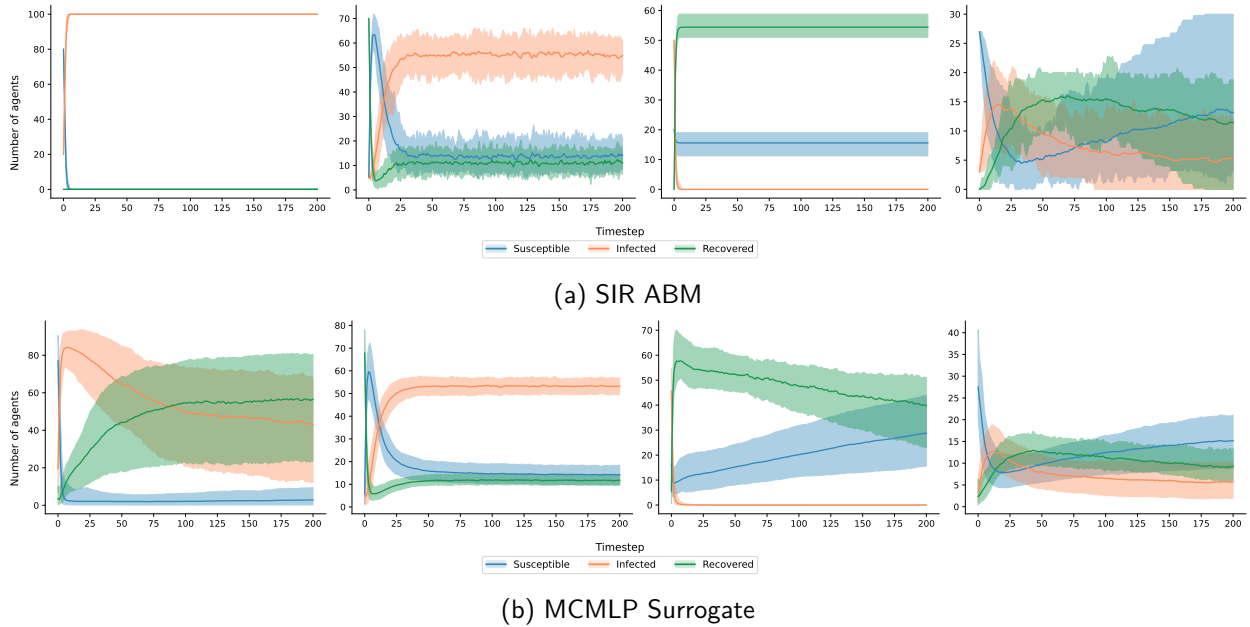


Figure 4.3: The MCMLP surrogate (b) struggles to capture the full range of behaviours of the SIR ABM (a)¹

Our intention was to develop surrogates that could capture most of the complexity that the SIR ABM exhibits. However, we quickly encountered significant difficulties. Despite scaling up MCMLP's hidden dimension $8\times$ and training on 400 000 examples, MCMLP struggled to mimic the ABM's behaviour across the parameter space. In Figure 4.3, we can see that MCMLP is largely unable to mimic the ABM's behaviour, either diverging over time or exhibiting totally different behaviour from the outset. Notably, the far left example in Figure 4.3, tests whether the surrogate has learnt that the ABM behaves as an SI model when $p_{\text{recover}} = 0$ and there are no initially recovered agents. Instead, the surrogate predicts a rapidly recovering population, something which should be impossible. These difficulties necessitated

¹Parameters from left to right: ($p_{\text{infect}} = 0.8$, $p_{\text{recover}} = 0.0$, $p_{\text{wane}} = 0.0$, $p_{\text{move}} = 0.0$, $\text{pop.} = 1.0$, $i = 0.2$, $r = 0.0$), ($p_{\text{infect}} = 0.3$, $p_{\text{recover}} = 0.1$, $p_{\text{wane}} = 0.5$, $p_{\text{move}} = 0.2$, $\text{pop.} = 0.8$, $i = 0.05$, $r = 0.7$), ($p_{\text{infect}} = 0.1$, $p_{\text{recover}} = 0.8$, $p_{\text{wane}} = 0.0$, $p_{\text{move}} = 1.0$, $\text{pop.} = 0.7$, $i = 0.5$, $r = 0.0$), ($p_{\text{infect}} = 0.6$, $p_{\text{recover}} = 0.05$, $p_{\text{wane}} = 0.025$, $p_{\text{move}} = 0.5$, $\text{pop.} = 0.3$, $i = 0.03$, $r = 0.0$)

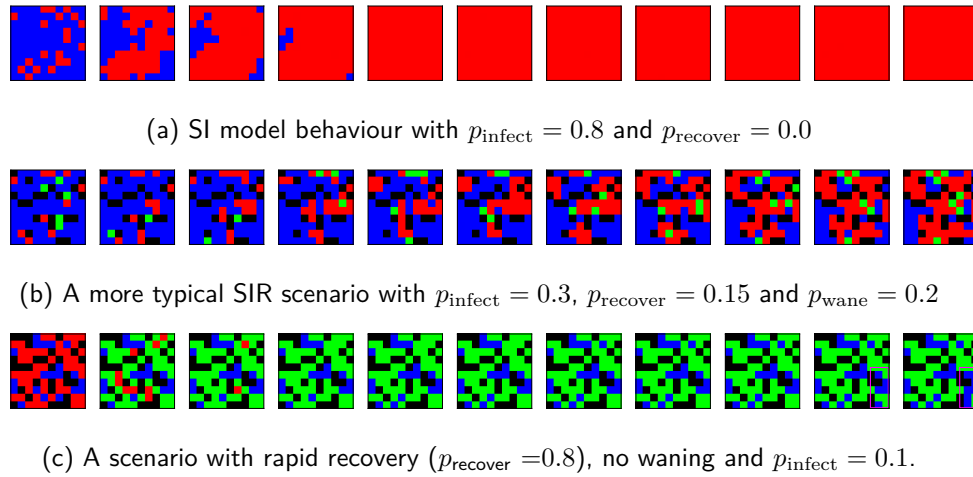


Figure 4.4: Example trajectories from the diffusion surrogate model. In (c) the magenta boxes indicate a region where recovered agents became susceptible again, which should not be possible.

that we aim to model only a subset of the ABM's full range of behaviours. We trained an MCMLP surrogate on ABM data that did not make use of agent movement (Figure A.17) and also trained another where there was no movement and a fully occupied grid (Figure A.16). Whilst there were some gains, both of these alternative models still totally diverged from the ABM in some cases.

Despite being trained on an image-based version of the same dataset, the diffusion surrogates were much better able to mimic the case where the ABM behaves as an SI model (See Figure 4.4a). This indicates that the image-based diffusion surrogates may be better able to learn the general rules of the ABM than surrogates which mimic count timeseries. However, even with the *colour snapping* mitigation detailed in Section 3.2.2, there are still hallucinations where unoccupied cells are filled or agent state transitions that should not be possible occur. In Figure 4.4c, $p_{\text{wane}} = 0$; however, we observe some recovered agents that become susceptible again in the region outlined by the magenta box. This should not be possible. Furthermore, we were unable to produce a diffusion surrogate that was faster than the original ABM. The average time taken by the diffusion surrogate to perform a single simulation step for a batch of 20 on a NVIDIA® GeForce® MX550 GPU was 0.011 seconds. In contrast, performing 20 simulation steps for a batch of 20 with the ABM took an average of 0.001 seconds. Therefore, in the time that the diffusion surrogate would take to perform 20 steps, the ABM's computation could be run 10 times. Our attempts to improve performance included massively scaling down the model, using depthwise separable convolutions (Sifre, 2014) to reduce the parameter count and attempting to predict several grid updates at once. Unfortunately, the performance requirements were too tight for this class of model, which typically requires sophisticated architectures to work well and needs iterative sampling by design.

We therefore turned our attention toward improving the MCMLP surrogate. By increasing the amount of data fivefold, doubling the hidden dimension of the model, training on trajectories of only 30 simulation steps, and disabling movement and sparse agent populations, the MCMLP surrogate was much better able to mimic the average realisations of the ABM (Figure 4.5). However, as before, the surrogates' aleatoric uncertainties typically did not closely match the ABM's.

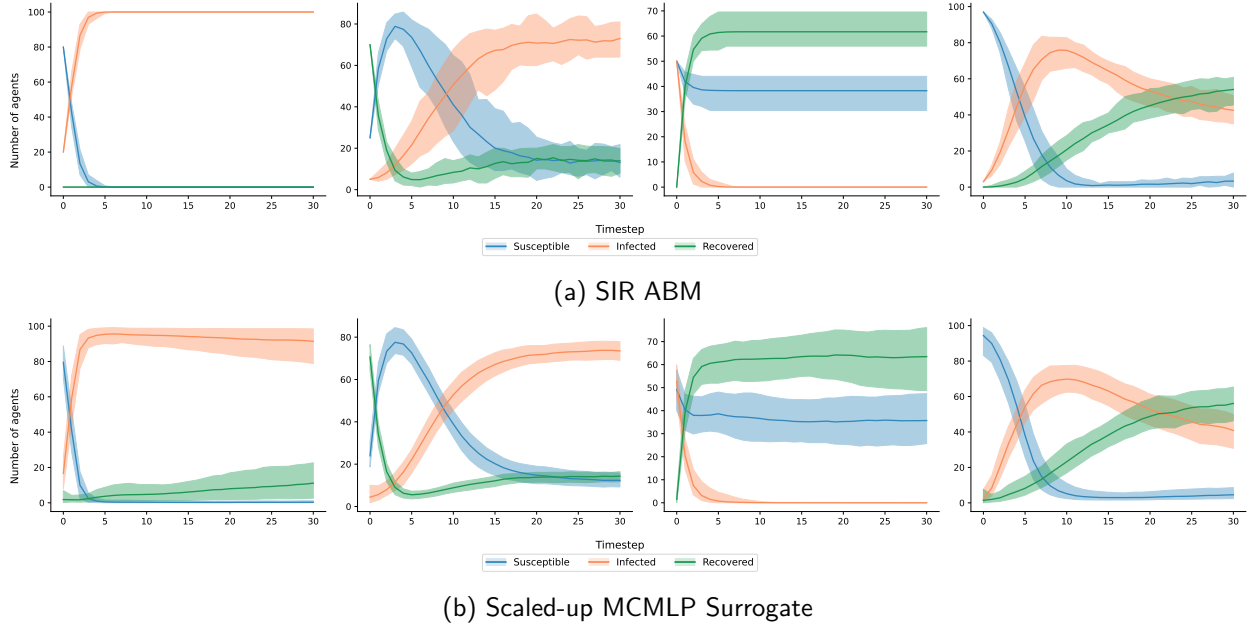


Figure 4.5: The scaled-up MCMLP surrogate (b) appears to more closely match the ABM (a)

4.2.2 Parameter Inference

Performing inference for our ABM with 200 simulation steps proved to be impractical as the dimensionality of the posterior made our early attempts at inference futile. Hence, we reduced the number of simulation steps to 30, incorporated a full covariance estimate as in (Wood, 2010), increased the number of ABM realisations per sample and allocated a sample budget of 300 000. For all these measures, none of our ABM inference tests fully converged ($\hat{r} \geq 1.1$) and all of them had a low number of effective samples. We were unfortunately not able to further iterate on our inference approach for the ABM, since inference for even a single example took an hour. Although inference did not converge, in Figure 4.6a, the marginal posteriors for p_{infect} and p_{wane} have most of their probability mass near the true parameter values, with p_{recover} not being as closely matched.

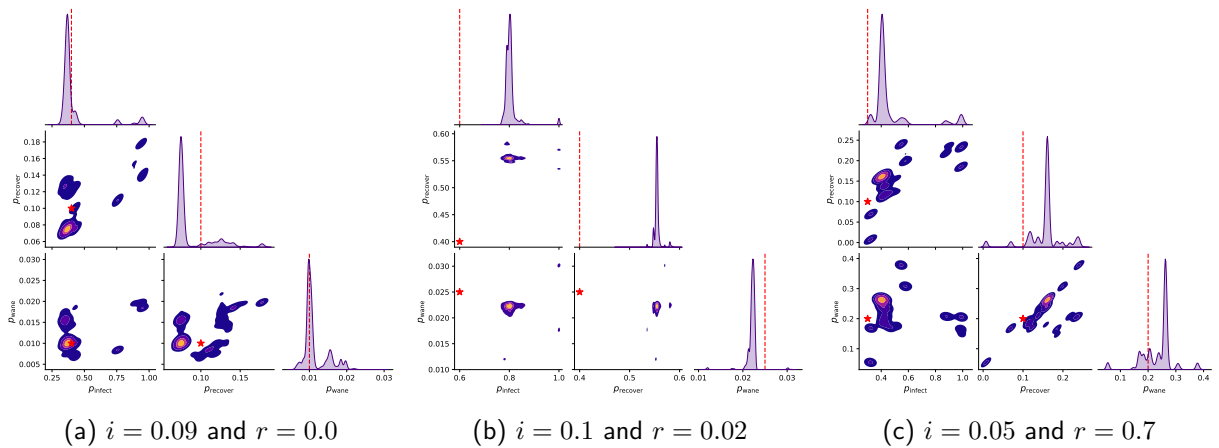


Figure 4.6: ABM posterior distributions for p_{infect} , p_{recover} and p_{wane} .

Adopting the same inference model for the surrogates produced even worse results. On all three inference

examples, the MCMLP surrogate's posterior did not deviate from the prior distribution. In order to verify that this issue wasn't specific to MCMLP, we trained a large PriorCVAE surrogate on the same data and observed the exact same phenomenon (Figure 4.8). Since both classes of surrogate suffer from this phenomenon and the MCMLP surrogate appears to capture the average behaviour of the ABM fairly accurately, we hypothesised that the issue might be related to the covariance estimate used by the likelihood. In Figure 4.5, the MCMLP's aleatoric uncertainty is consistently different to the ABM's, and this may lead to covariance estimates that substantially differ. In order to better understand this phenomenon, we performed inference for the MCMLP surrogate with a fixed diagonal covariance.

Table 4.3: Comparison of Negative Log Predictive Density (NLPD) and inference time between the ABM and MCMLP with fixed covariance.

	NLPD		Inference Time (s)	
	ABM	MCMLP	ABM	MCMLP
Example A	1.217e+02	3.631e+03	3248	914
Example B	NaN	3.308e+03	3219	914
Example C	1.576e+02	2.193e+03	3231	912

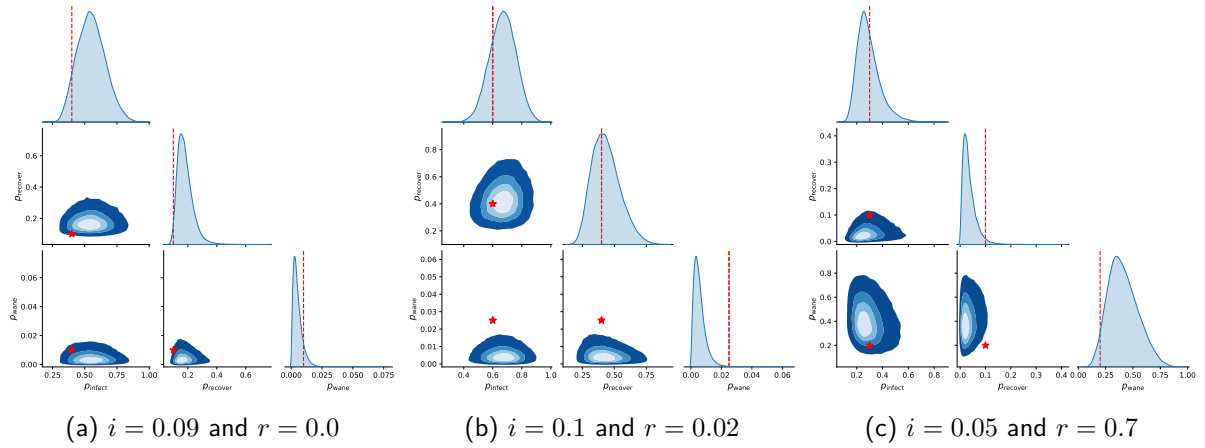


Figure 4.7: By fixing the covariance of the likelihood to $5\mathbf{I}$, inference with the MCMLP surrogate is possible.

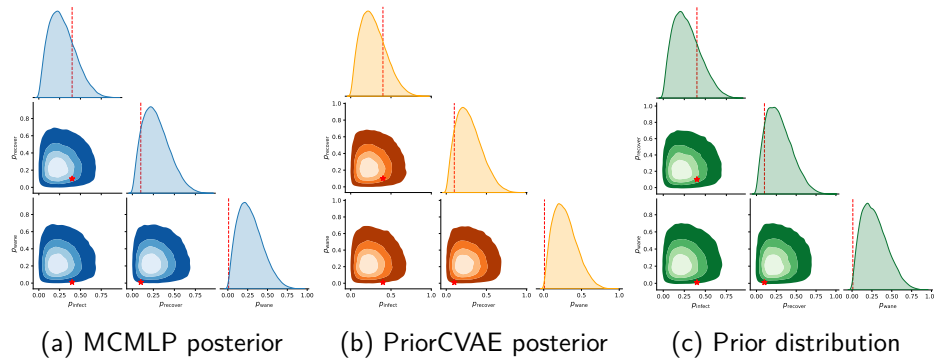


Figure 4.8: The posterior distributions of the surrogates when estimating the full covariance are unchanged from the $\text{Beta}(5, 2)$ prior on p_{infect} , p_{recover} and p_{wane} .

The use of a fixed diagonal covariance enabled parameter inference (Figure 4.7); however, in two of the three test cases, the ABM's NLPDs were better by an order of magnitude (Table 4.3). In the other test case, Example B, the ABM's NLPD was *Not a Number* (NaN). This is likely due to the ABM's posterior being concentrated far from the true parameter values (Figure 4.6b).

5. Discussion

In this work, we have investigated the challenges associated with developing surrogates for Agent-Based Models. In particular, we focused on the task of parameter inference from data, which is both essential for the application of a computational model and difficult to do with an ABM.

Utility of lightweight surrogates for simple ABMs

An essential step before developing surrogates for ABMs is optimising the ABM itself, since this step may remove the need for a surrogate altogether. We were able to make a simple SI ABM adapted from (Semenova, 2024) over $600\times$ faster at the largest grid size we profiled by replacing conditional logic with convolutions and elementwise comparisons. However, even with these optimisations, we found that performing parameter inference for the ABM quickly became impractical as the size of the grid grew. Experiments with lightweight surrogates proved fruitful; a small Monte-Carlo Dropout (MCMLP) surrogate, which trains in under a minute on a modest dataset of samples, accelerated parameter inference by orders of magnitude whilst yielding predictive performance comparable to the ABM itself. These speed-ups make it feasible to integrate this surrogate into a full Bayesian workflow (Gelman et al., 2020), where the parameter inference process is iteratively refined and where the cost of training would be amortised.

Sample efficiency and fidelity trade-offs

Following our initial experimentation we developed a Susceptible-Infected-Recovered (SIR) ABM that is more representative of the complexity found in the ABMs that are applied in practice. In addition to modelling the spread of disease between agents, it can simulate agents obtaining immunity through recovery from the disease and the waning of that immunity. Furthermore, it allows for the distribution of agents on the grid to be sparse and for agents to move into unoccupied cells. The increased complexity of this ABM made it challenging for our low-fidelity surrogates to model the range of behaviours that the SIR ABM can exhibit. This necessitated the creation of a larger dataset, disabling agent sparsity and movement, and only training on shorter realisations. These challenges motivated the concurrent exploration of a high fidelity diffusion-based surrogate which directly predicts the states of the ABM simulation. This surrogate proved to be far better at modelling the SIR ABM at a similar parameter count and demonstrated greater sample efficiency. However, it proved to be impractical as it was significantly slower to sample from than the ABM, despite our best efforts to optimise it.

Interaction between the synthetic likelihood and surrogates

The synthetic likelihood (Wood, 2010) requires estimates of the mean and covariance of realisations generated by the ABM or surrogate. We found that even when surrogates accurately capture the mean behaviour of the ABM, an inaccurate model of the stochastic properties of ABM makes parameter inference with a synthetic likelihood challenging. PriorCVAE's consistent underestimation of the aleatoric uncertainty of the SI ABM led to underestimated parameter and predictive uncertainty. These challenges became far more severe when performing inference for the SIR ABM. Employing a surrogate likelihood with covariances estimated from either MCMLP or PriorCVAE surrogates caused inference to fail to the extent that the posterior did not move from the prior. Switching to a constant diagonal covariance

matrix allowed the inference to proceed, suggesting that an overly simplistic likelihood may be more effective than a more expressive likelihood which is highly sensitive to errors in its estimation.

Conclusion and Future Work

Our work demonstrates that whilst deep generative surrogates offer great promise in making Agent-Based Models more widely applicable, their application is not straightforward. As the complexity of the ABM increases, a trade-off emerges between low-fidelity surrogates that require a large number of examples to capture complex dynamics and high-fidelity surrogates which have greater sample efficiency but which may be slower than the ABM they are meant to replace. The increase in complexity also makes the covariance estimate used in a synthetic likelihood more critical, as imperfections in the surrogate's estimation of variance-covariance structure may lead to non-convergence. For this reason, we believe that future work would do well to carefully investigate design choices around the surrogate model's likelihood as well as thoroughly explore advances in likelihood-free inference.

Although we were able to produce an MCMLP surrogate that accelerated parameter inference for a limited version of the SIR ABM, training this surrogate required a large number of samples, something that would prove to be impractical for larger grid sizes or more complex ABMs. Furthermore, the NLPDs of this surrogate were significantly worse than the ABM's. We believe that integrating techniques from Active Learning and Active Sampling may help to address the sample efficiency concerns and that improvements to the formulation of the surrogate's likelihood, or likelihood-free inference could improve predictive performance.

For ABMs that are highly complex and difficult to optimise further, we believe that high fidelity diffusion-based surrogates may still prove useful, as their greater sample efficiency may make them more practical as data generation for surrogate training becomes a bottleneck. In this vein, Discrete Flow Matching models (Gat et al., 2024) are an interesting avenue to explore, since they will not require the tricks we employed to use diffusion with discrete agent states.

Code Availability

The code to reproduce our work is available at https://github.com/SteamedGit/aims_dgs4abm

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Appendix A.

A.1 Susceptible Infected ABM Methods

Table A.1: Hyperparameter search spaces for MCMLP and PriorCVAE.

Hyperparameter	Values/Range
MCMLP	
hidden_dim	{64, 128, 256}
learning_rate	$[10^{-6}, 10^{-2}]$
n_epochs	{100, 200}
PriorCVAE	
encoder_hidden_dim	{128, 256, 512}
encoder_num_layers	{3, 4}
decoder_hidden_dim	{128, 256}
decoder_num_layers	{2, 3, 4}
learning_rate	$[10^{-6}, 10^{-2}]$
n_epochs	{100, 200}

A.2 Susceptible Infected ABM Results

A.2.1 Parameter Estimation and Posterior Prediction

In this section we present the results for every parameter inference and posterior prediction evaluation performed across the different grid and simulation step configurations for the SI ABM. The parameter inference results are annotated with the $\hat{r}$ values achieved by the respective methods, and hatching is used to visually indicate when a method's $\hat{r}$ was 1.1 or greater. The posterior prediction figures are annotated with each method's NLPD calculated on 1000 groundtruth examples from the ABM. The data used to perform parameter inference is also plotted for completeness.



Figure A.1: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 10 with 10 simulation steps.

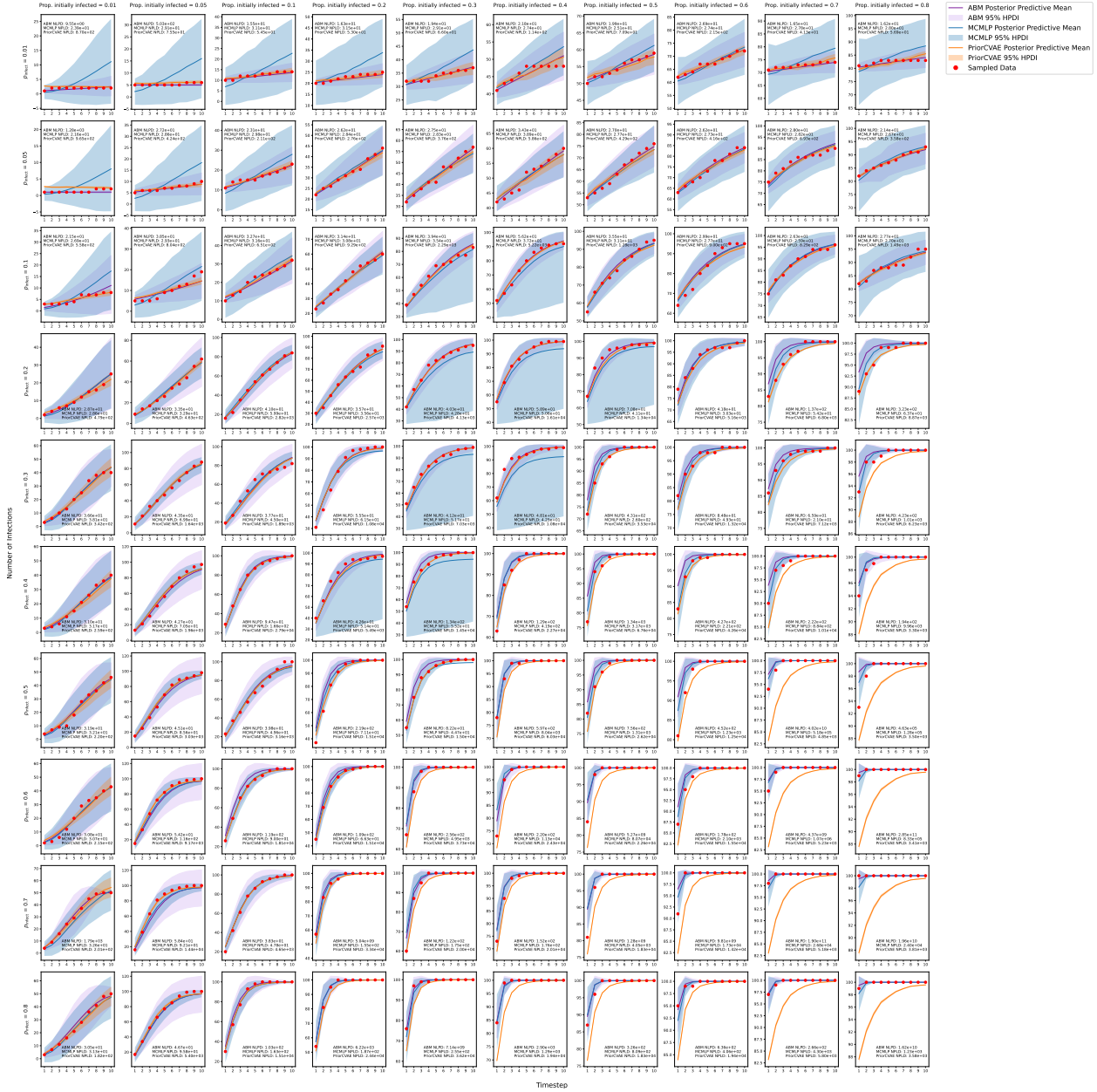


Figure A.2: ABM and Surrogate posterior predictions at grid size 10 with 10 simulation steps.



Figure A.3: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 10 with 20 simulation steps.

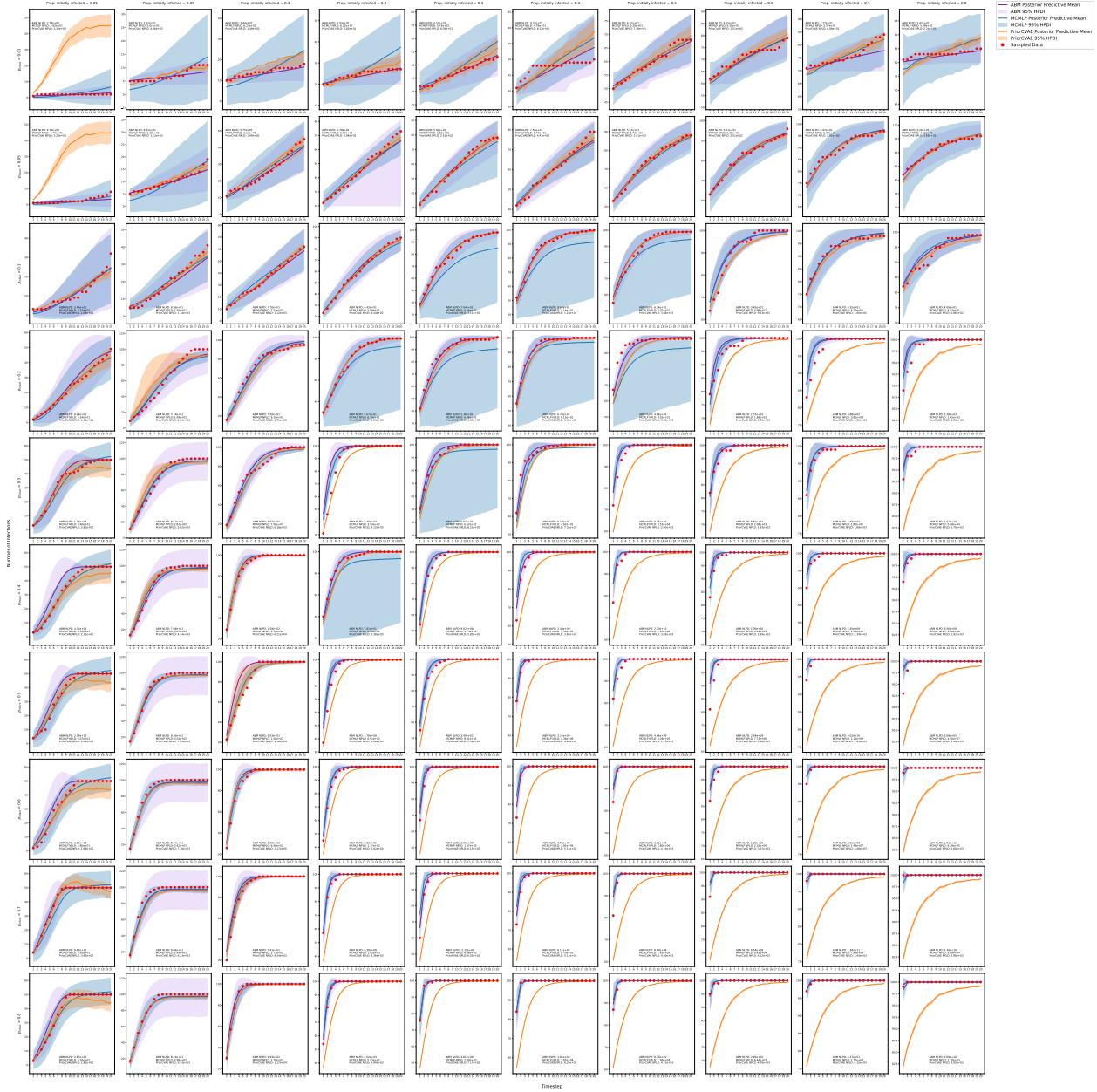


Figure A.4: ABM and Surrogate posterior predictions at grid size 10 with 20 simulation steps.

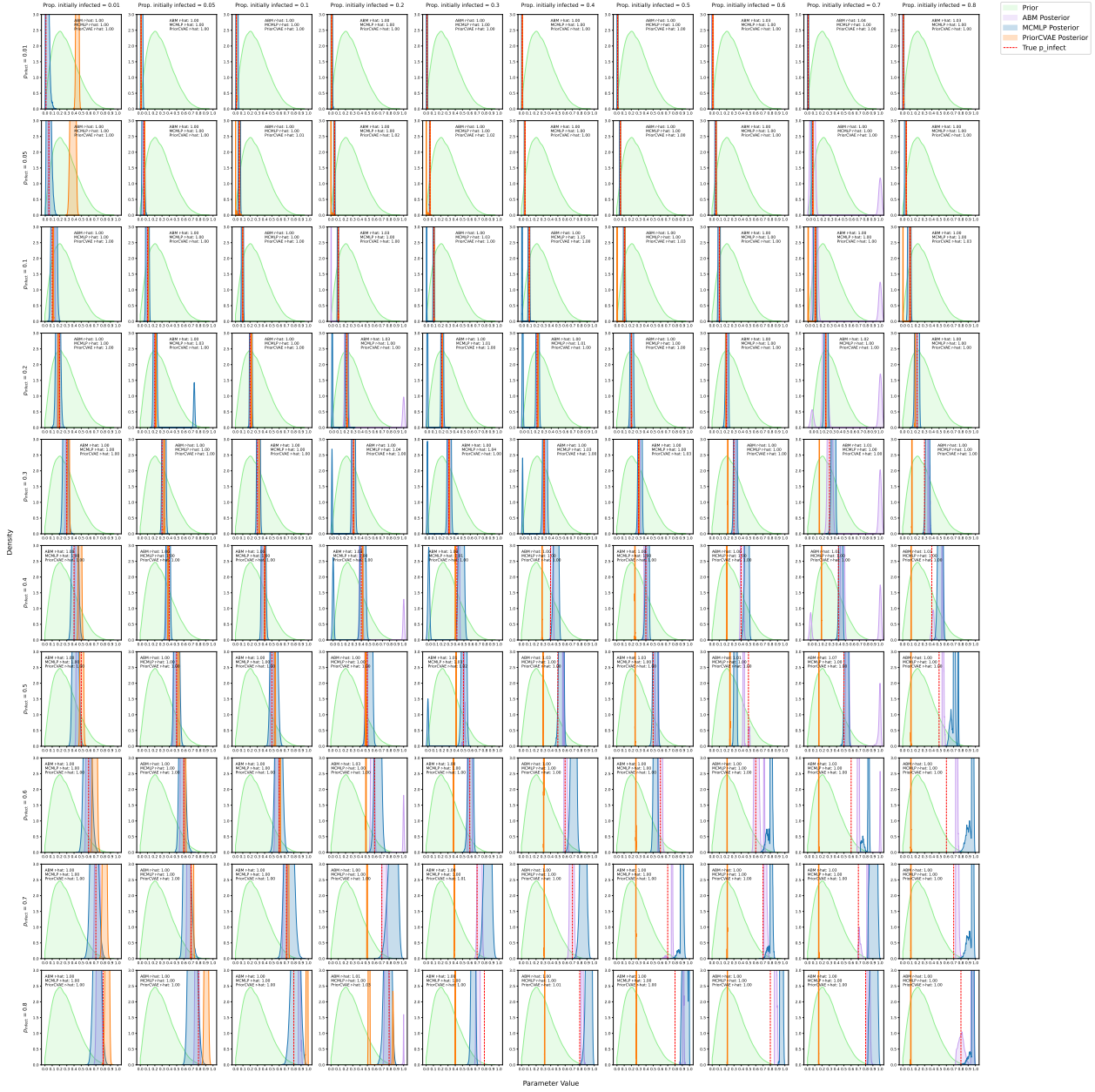


Figure A.5: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 40 with 10 simulation steps.

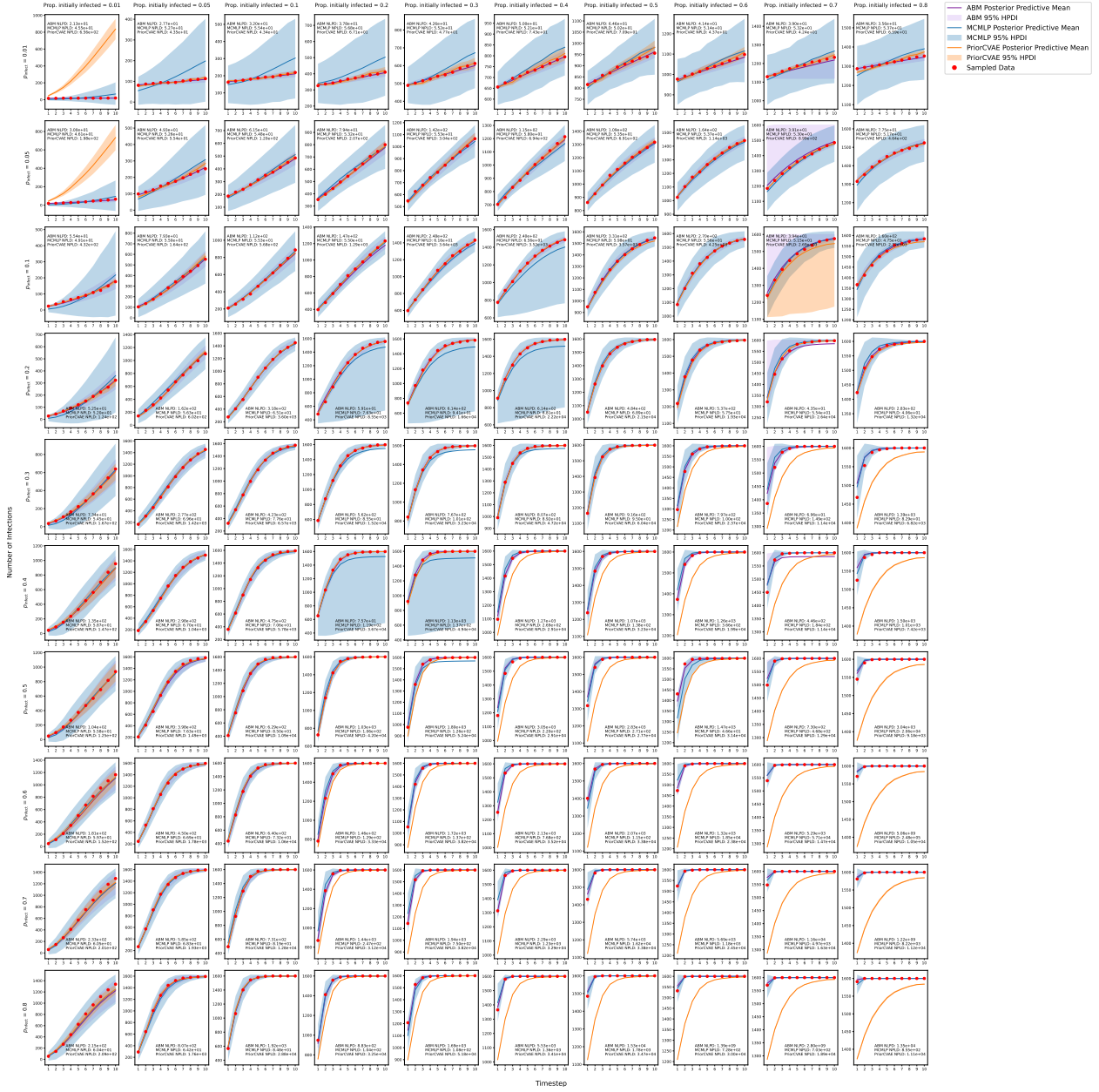


Figure A.6: ABM and Surrogate posterior predictions at grid size 40 with 10 simulation steps.



Figure A.7: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 40 with 20 simulation steps.

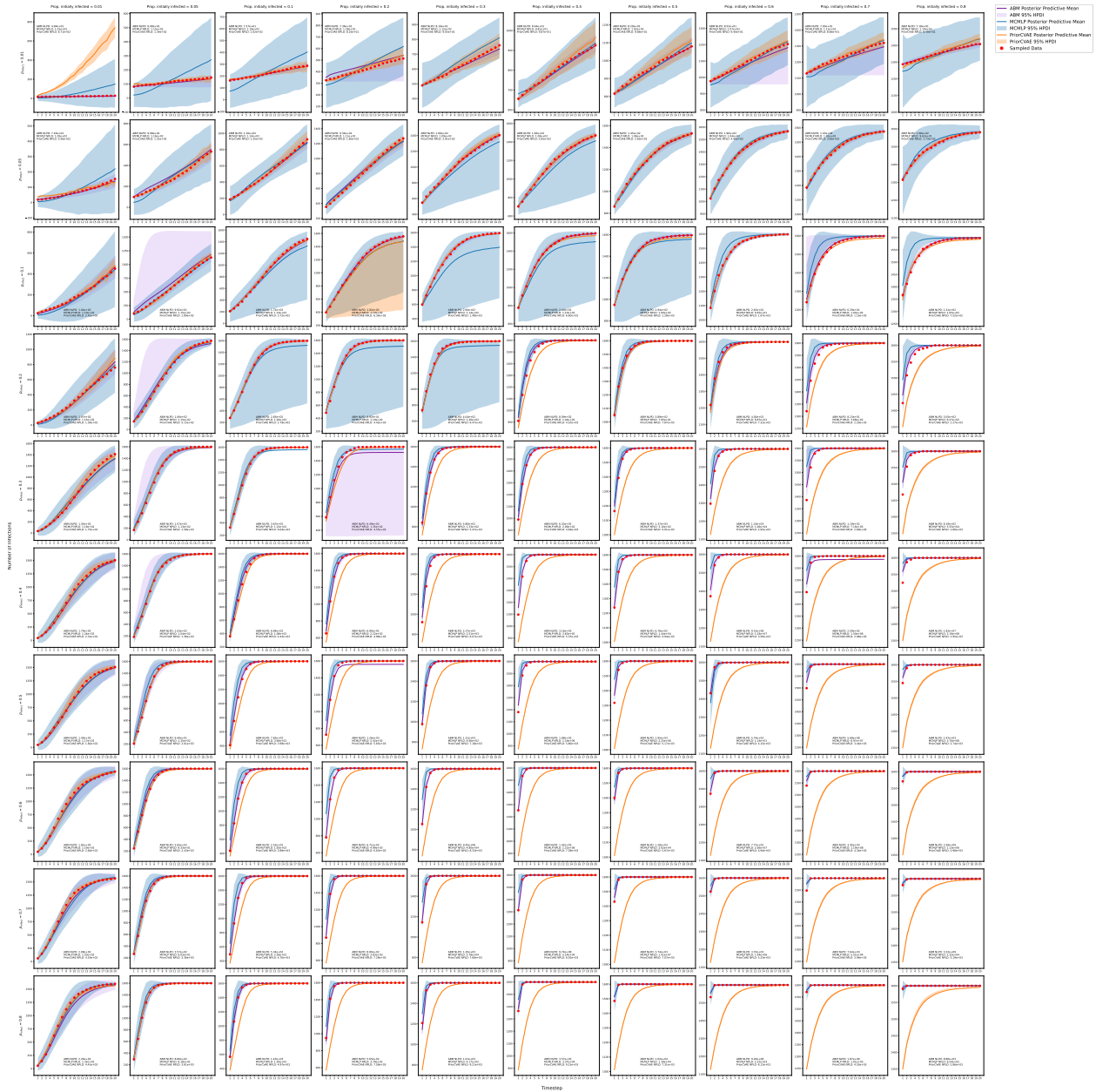


Figure A.8: ABM and Surrogate posterior predictions at grid size 10 with 20 simulation steps.

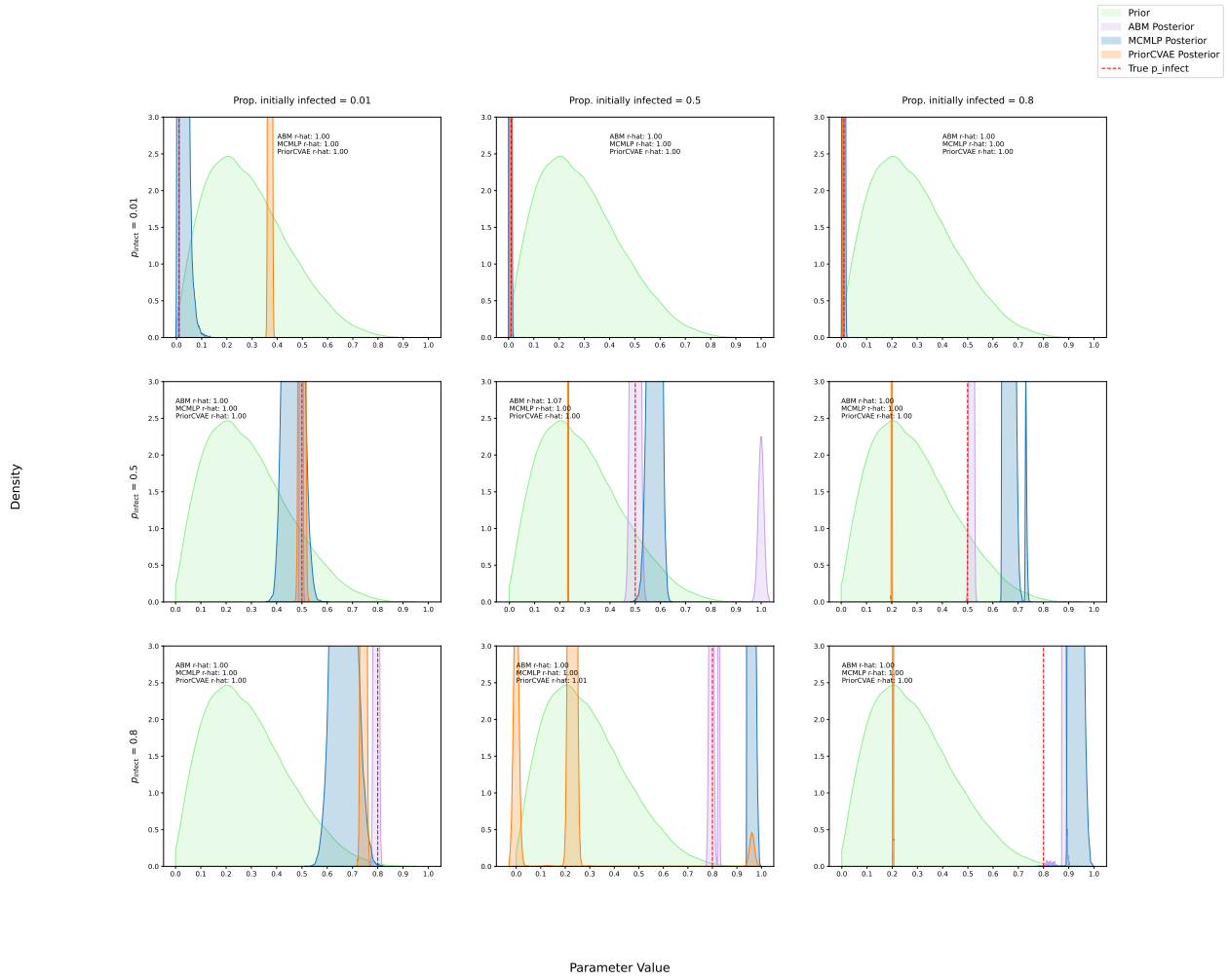


Figure A.9: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 160 with 10 simulation steps.

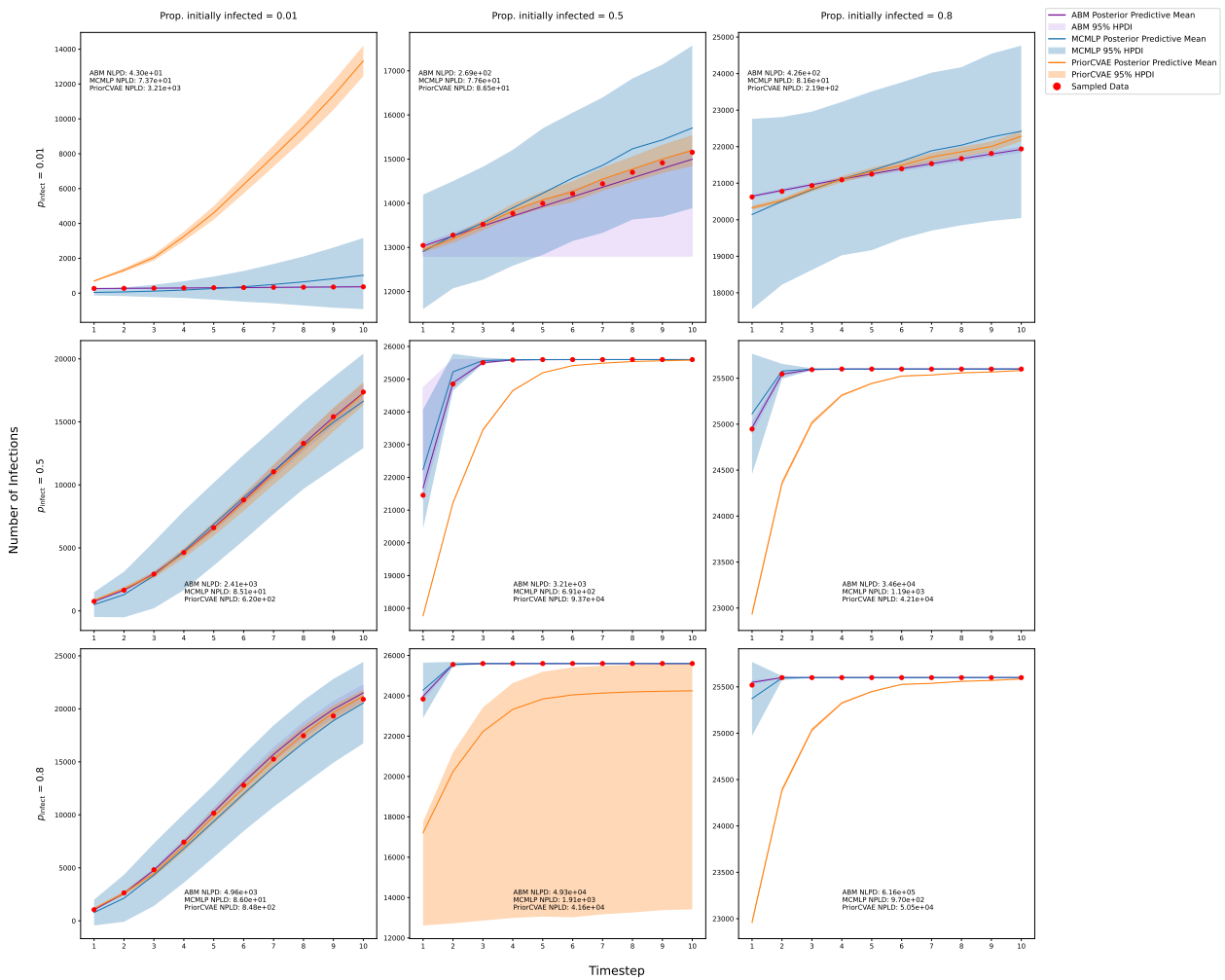


Figure A.10: ABM and Surrogate posterior predictions at grid size 160 with 10 simulation steps.

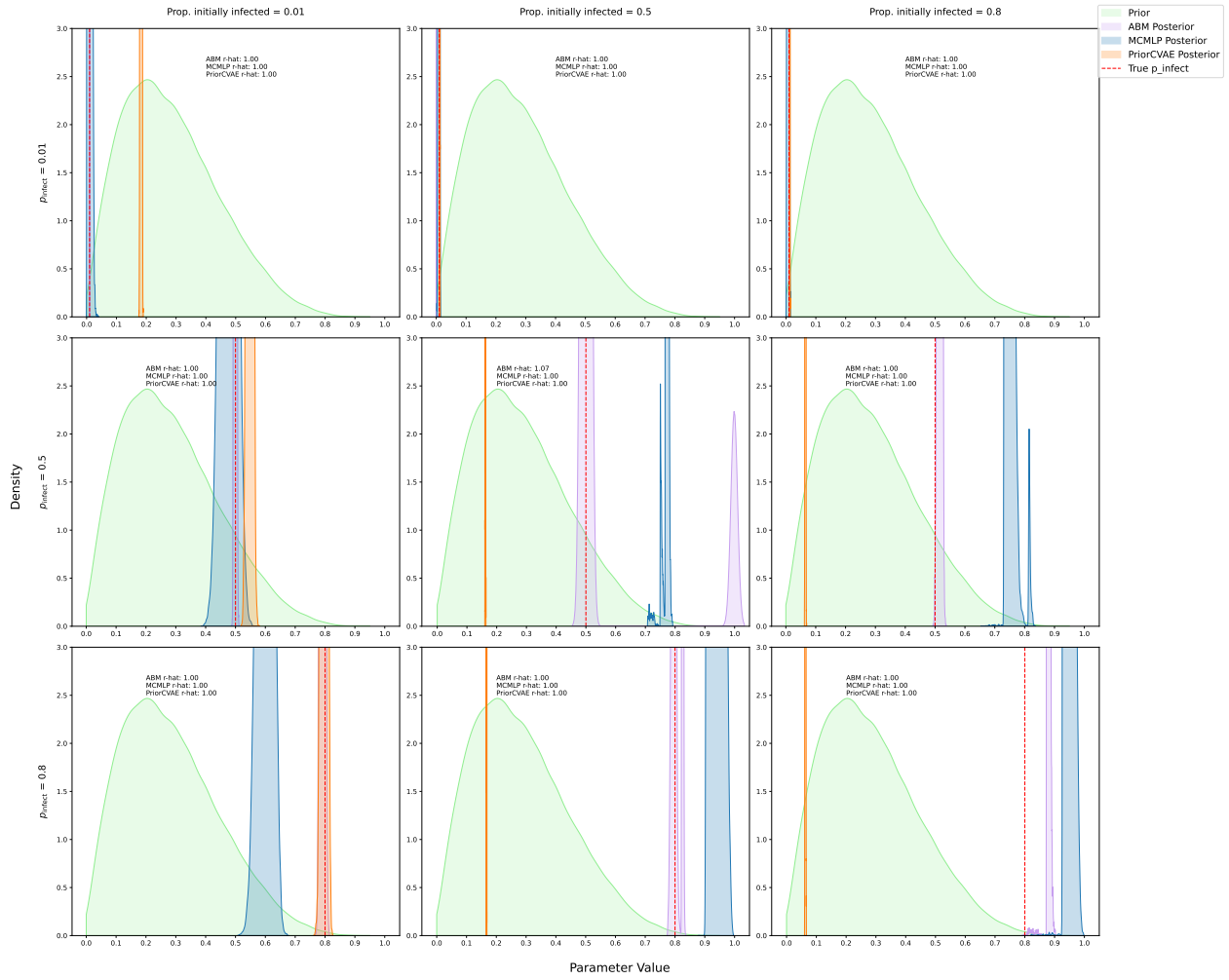


Figure A.11: ABM and Surrogate posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 160 with 20 simulation steps.

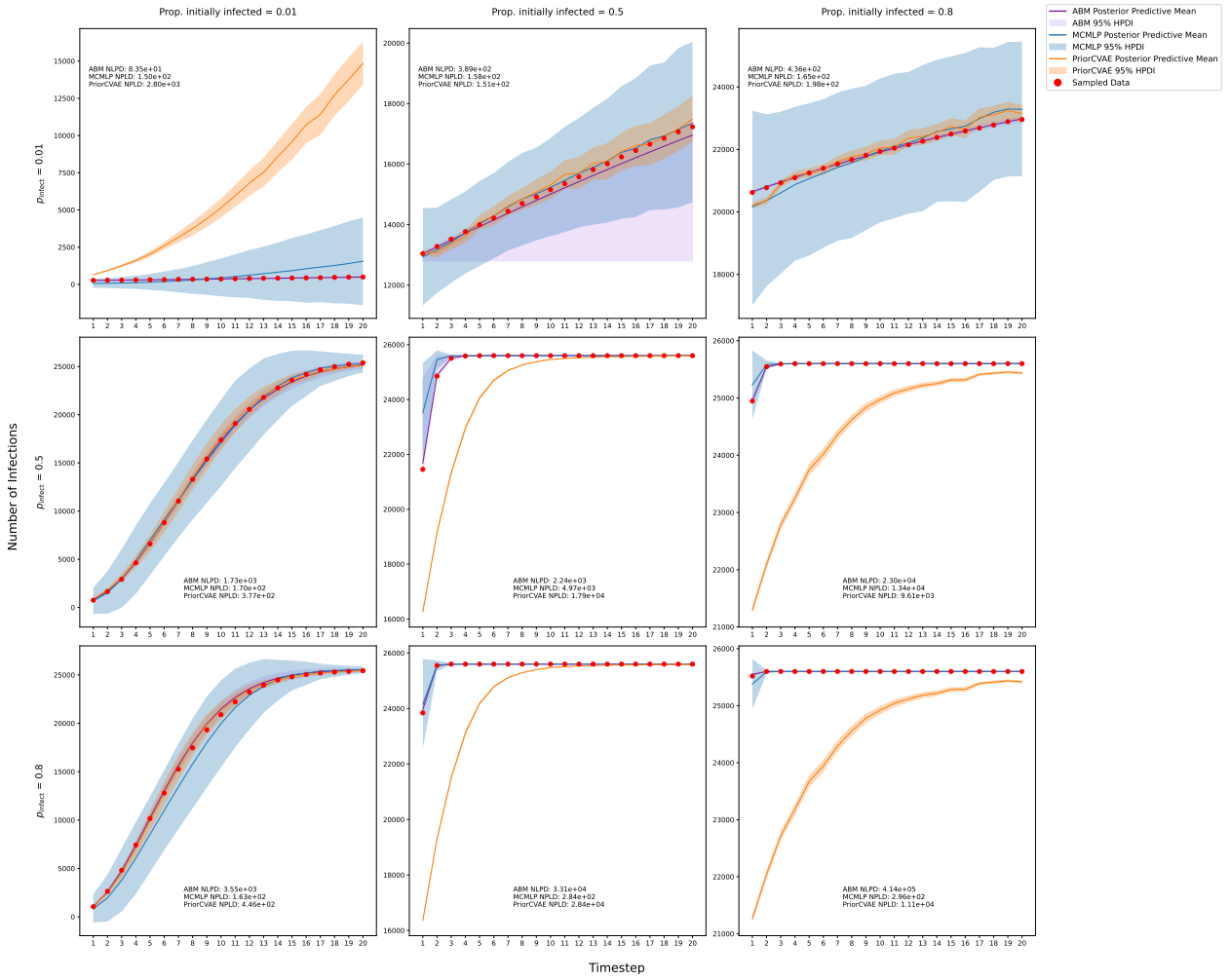


Figure A.12: ABM and Surrogate posterior predictions at grid size 160 with 20 simulation steps.

A.2.2 Experiments with varying PriorCVAE's σ_{vae}^2

As the results obtained with the PriorCVAE surrogate consistently underestimated parameter and predictive uncertainty we experimented with increasing σ_{vae}^2 to determine if this could yield stronger results. Given the optimal configuration on a 40×40 grid and 10 simulation steps, we trained PriorCVAE variants with $\sigma_{\text{vae}}^2 = 2$, $\sigma_{\text{vae}}^2 = 5$, $\sigma_{\text{vae}}^2 = 10$. Unfortunately, these models appeared to have learnt near-constant functions which overlap in Figure A.13 and Figure A.15. Unsurprisingly, these variants performed worse in every metric compared to the original PriorCVAE surrogate (Table A.2).

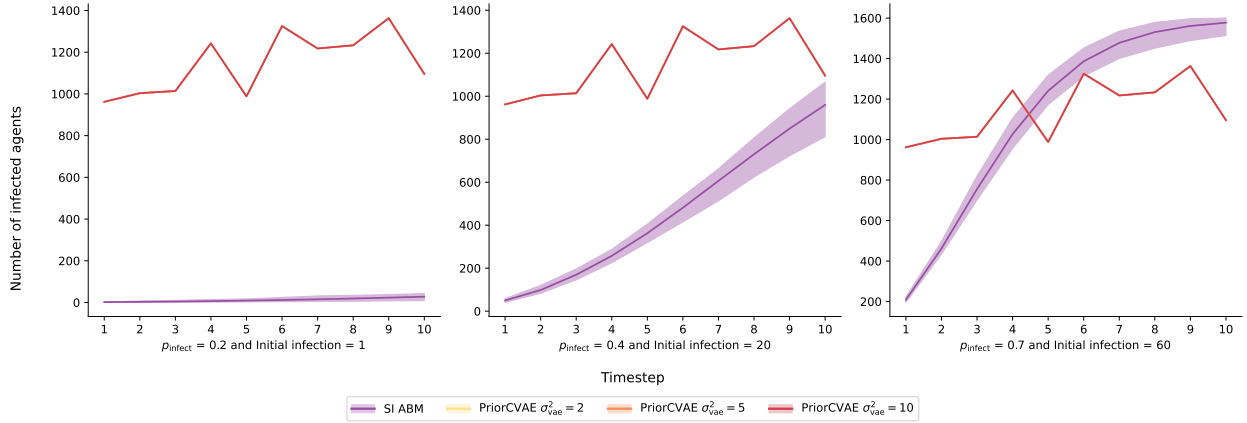


Figure A.13

Table A.2: Comparison between the original PriorCVAE surrogate for a 40×40 SI ABM running for 10 simulation steps and variations which increase σ_{vae}^2 .

Metric	$\sigma_{\text{vae}}^2 = 1$ (default)	$\sigma_{\text{vae}}^2 = 2$	$\sigma_{\text{vae}}^2 = 5$	$\sigma_{\text{vae}}^2 = 10$
Posterior Means MSE	0.066	0.187	0.190	0.189
Mean W1	0.155	0.378	0.380	0.379
NLPDs	8/100	4/100	4/100	4/100

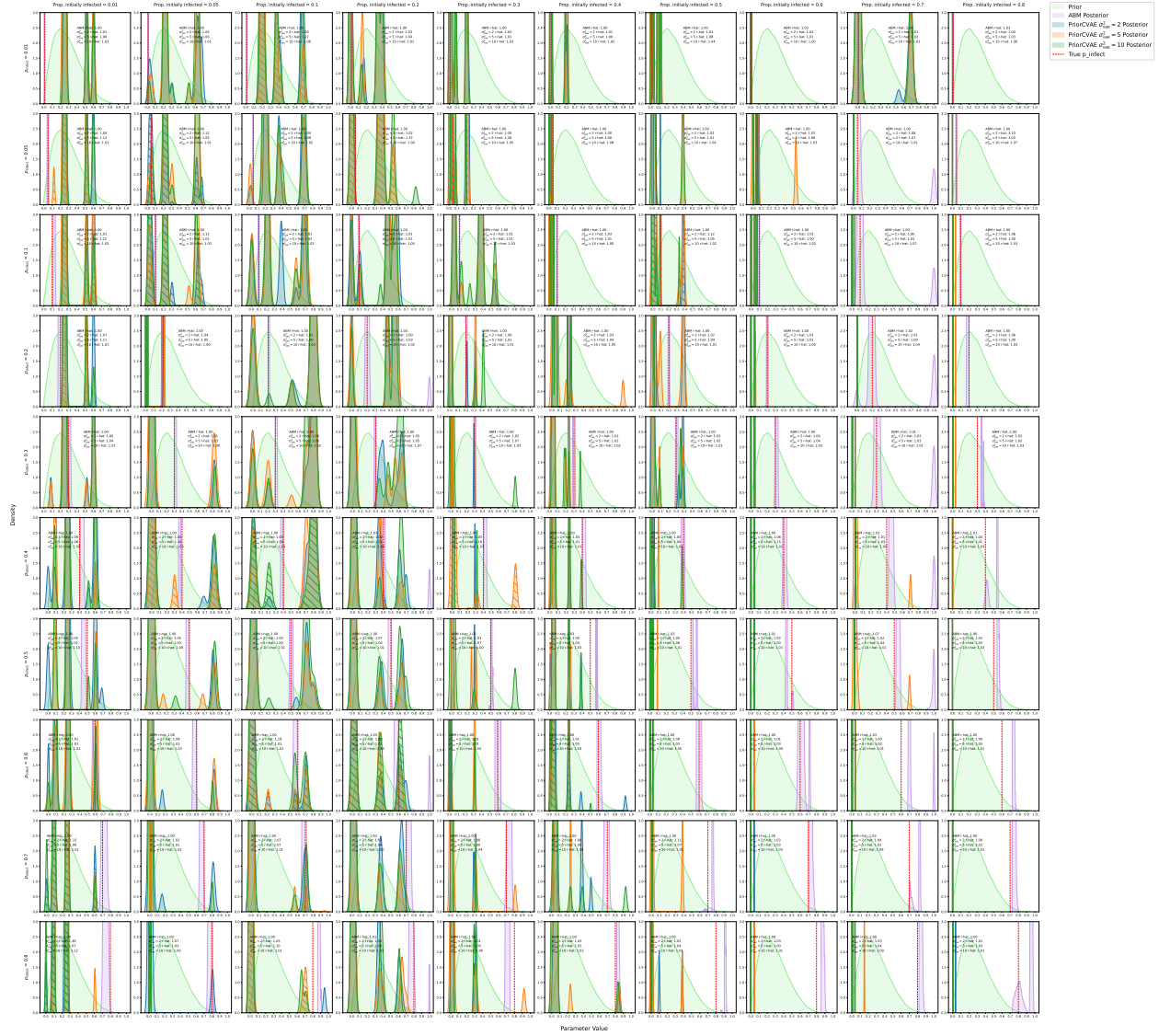
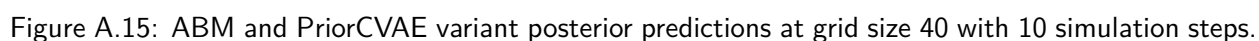


Figure A.14: ABM and PriorCVAE variant posteriors across different combinations of probability of infection and sizes of initially infected populations at grid size 40 with 10 simulation steps.



A.3 Susceptible Infected Recovered ABM Results

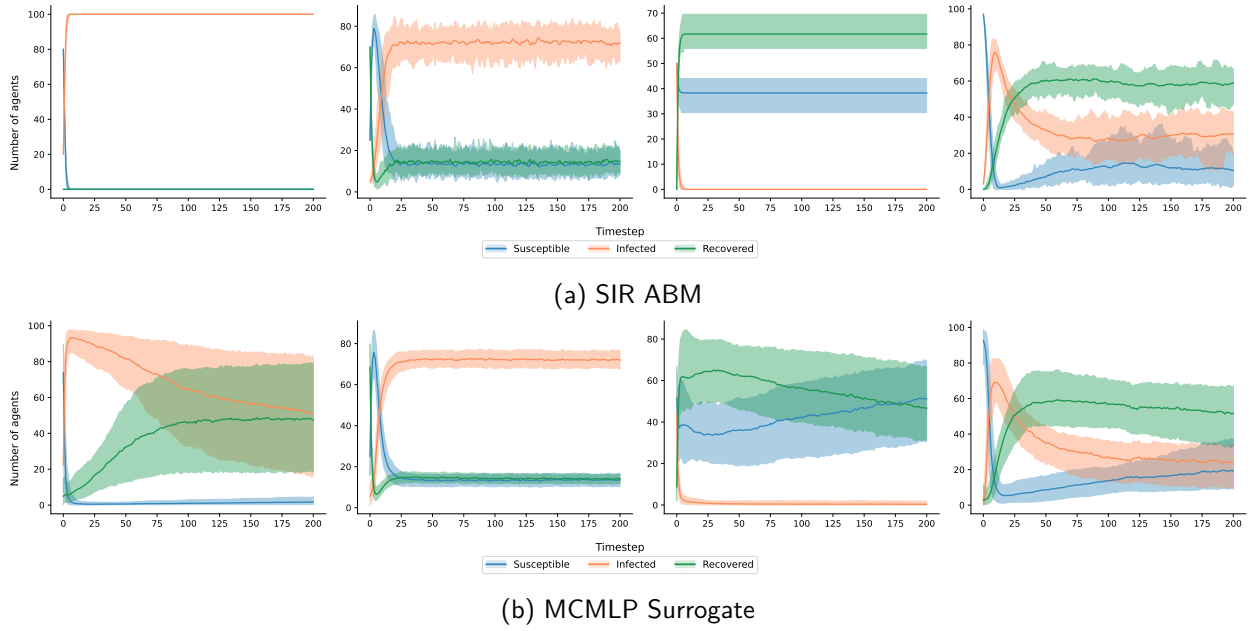


Figure A.16: Comparing an MCMLP Surrogate trained on SIR data with a fully occupied grid and no movement with the SIR ABM.

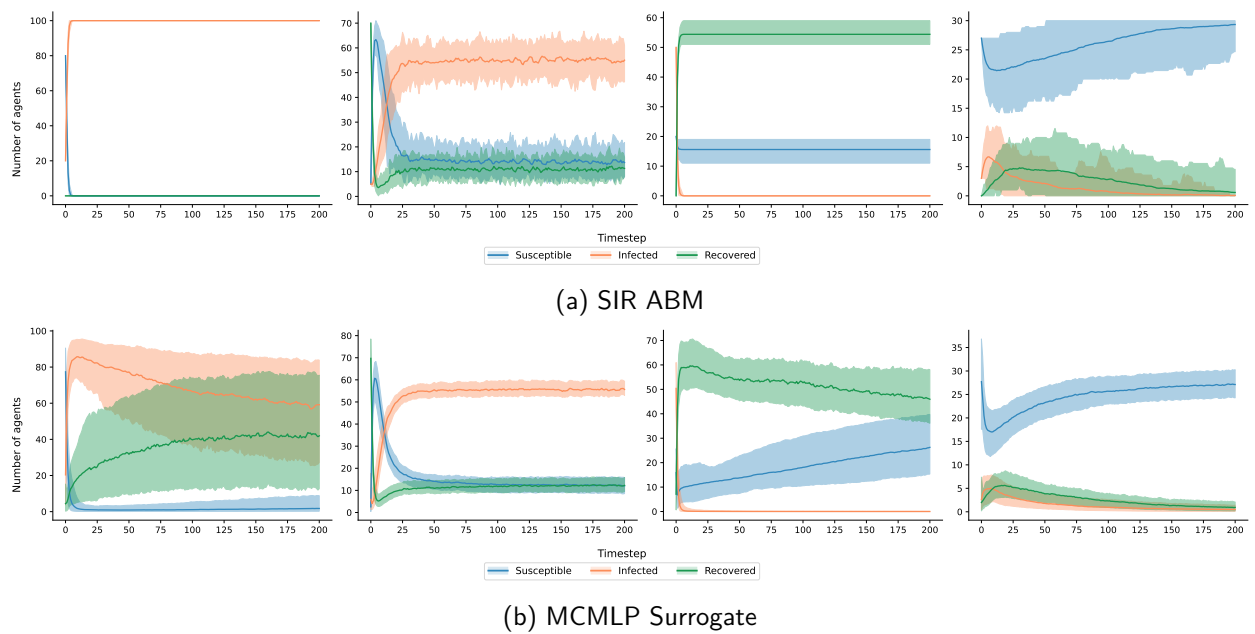


Figure A.17: Comparing an MCMLP Surrogate trained on SIR data with no movement with the SIR ABM.